

The United States as the International Lender of Last Resort

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Abstract

This paper provides a stylized framework to study the role of the United States as the international lender of last resort to global banks. The model captures a central feature of the international financial system, namely, non-US global banks that invest heavily in US assets but are exposed to dollar liquidity shortages. This situation can give rise to multiple equilibria, one of which resembles a global financial crisis, with a sharp appreciation of the dollar, tighter financial conditions in international markets, weaker global economic activity, and struggling banks. The self-fulfilling nature of the crisis stems from a feedback loop between the exchange rate and the capacity of non-US banks to raise funds. Since the liquidity needs of these banks are often denominated in dollars, the Federal Reserve is better equipped than other central banks to prevent the “bad” equilibrium when the dollar is strong. However, its incentives to intervene—through swap lines—may not be aligned with the rest of the world because of general equilibrium forces that drive cheaper capital flows into the US during times of global financial stress.

JEL Codes: F33, F41, E44

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1 Introduction

In banking, a common practice is to finance long-term assets with short-term liabilities, which can expose banks to liquidity shortages. To prevent the amplification of this risk throughout the economy, a classic solution is to have a lender of last resort—typically the domestic central bank—that provides the liquidity needed in times of stress ([Bagehot, 1873](#)). Recent decades, however, have been marked by the rise of large global banks that operate in multiple regions and engage in maturity transformation on a global scale. Importantly, most of their cross-border transactions are denominated in dollars, even though many of these banks are non-US intermediaries. This dollar dominance poses difficulties for domestic central banks in covering the short-term needs of their banks during times of stress, when liquidity is scarce and the dollar appreciates sharply.

In response, the United States has increasingly assumed the role of the *international lender of last resort*. Since the 2008 global financial crisis (GFC), the Federal Reserve has provided dollar liquidity to foreign central banks—primarily through swap lines¹—allowing them to support their domestic financial systems. While this intervention has become a central pillar of the international financial architecture ([Bahaj and Reis, 2022b](#)), many fundamental questions remain: What are the macroeconomic effects of swap lines? How does the role of an international lender of last resort differ from a domestic one? And are the incentives of the United States always aligned with those of the rest of the world?

This paper presents a framework that rationalizes the United States’ provision of dollar liquidity to the world, and analyzes its macroeconomic implications. I argue that a world populated by non-US global banks with dollar maturity mismatches is prone to self-fulfilling crises driven by a two-way interaction between the exchange rate and banks’ funding constraints. During such episodes, the dollar endogenously appreciates and domestic central banks struggle to provide the dollar liquidity needed to stabilize their banking systems. In contrast, the Fed could intervene—but it may not fully internalize the benefits of such intervention for the world, since the US enjoys cheaper capital inflows and a stronger currency during periods of global financial stress.

To formalize this insight, I develop a stylized model of the world economy that captures important features of the modern financial system. I combine elements from the traditional self-fulfilling crises² literature with a modern perspective that places non-US global banks at the center of the international financial intermediation. This allows me to explore two understudied dimensions of the global financial system: the exposure of

¹In short, a swap line is an agreement between two central banks to exchange currencies at a specific exchange rate and for a limited period of time. [Section 2](#) provides more details about this instrument.

²This type of frameworks have been used to study mostly emerging markets, and were particularly relevant to understand the financial crises that they faced during the 90’s.

advanced economies to dollar fluctuations stemming from maturity mismatches, and the subsequent international spillovers from this exposure to other advanced economies. Both are crucial to understand the role of the Fed in providing dollar liquidity to the world. In addition, my framework highlights the importance of general equilibrium forces in explaining the appreciation of the dollar during turbulent times, and the heterogeneous impact of global crises on different economies.

I consider a world composed by two economies—the US and the euro area (EU)—populated by households and financially constrained global banks owned by EU households. These banks need to roll over their short-term debt in both euros and dollars to remain operational and benefit from the returns on long-term illiquid investment projects in both regions. However, they may fail to do so due to an agency problem: a stronger dollar increases the burden of their short-term dollar liabilities, while the friction prevents them from pledging sufficient future dollar asset returns to secure refinancing. As a result, even though these intermediaries are solvent in dollars, their borrowing constraint tightens, potentially leading to liquidity shortfalls and the fire sale of long-term assets. The resulting losses in the banking sector depress aggregate demand in the EU and may further reinforce the dollar appreciation.

The feedback loop between the exchange rate and the soundness of global banks opens the door to multiple equilibria in the spirit of [Bocola and Lorenzoni \(2020\)](#). In one equilibrium, the dollar remains relatively weak, banks successfully intermediate capital flows, and their long-term investments mature. The other equilibrium, on the contrary, resembles a global financial crisis, characterized by a sharp appreciation of the dollar, tighter conditions in financial markets, an increase in capital flows towards the US, and lower aggregate demand in the rest of the world.

Interestingly, self-fulfilling expectations about the exchange rate can trigger such crises. If agents are pessimistic and expect a significant depreciation of the euro, they anticipate tighter financial constraints for non-US banks. As a result, they withhold funding, prompting a costly liquidation of the banks' long-term assets. These losses lead to a decline in aggregate demand in the EU that is ultimately accommodated by a euro depreciation, validating the initial pessimistic expectations. These crises resemble not runs on individual institutions (as in [Diamond and Dybvig, 1983](#)), but systemic runs on the entire banking sector. In this sense, fundamentals matter: I show that crises are likely to arise only when short-term dollar liabilities are large, financial conditions are tight, and interest rates are high.

Next, I study the role of governments or central banks in preventing a crisis. The particular intervention I consider is in the form of a lender of last resort. In my framework crises occur due to pessimistic expectations, preventing households from providing the funding that banks need to operate. If a central bank can credibly commit to supplying liquidity, even when private sector expectations are pessimistic, it can prevent the crisis.

However, in a state of the world where the dollar is strong relative to other currencies, and since the liquidity needs of global banks are in dollars, non-US central banks without significant foreign currency reserves might lack the resources to prevent the financial crisis. Given its broad access to dollar liquidity, the Fed is better equipped to perform such an intervention, which can also be interpreted as a “bailout” for foreign banks.

Finally, I analyze the incentives for both economies to intervene. Indeed, while both the US and EU suffer from the collapse of global banks—through lost investment, partial loss of deposits, and disrupted financial flows—general equilibrium forces complicate the picture. A global crisis leads to capital inflows into the US and a stronger dollar, which boosts its relative wealth and tradable consumption. This mechanism resembles a scenario where the US is considered a safe haven during periods of stress. If these benefits outweigh the direct costs, the Fed may have little incentive to provide international liquidity. One interpretation of these findings is that the trade-off that the US faces could result in an underprovision of dollar liquidity³ to the world.

In the Online Appendix I also provide several extensions in which I explore different specifications for the main parts of the model. Most importantly, I use a three-period version of the baseline model to endogeneize the funding and investment decisions of non-US global banks. I show that, despite banks can choose ex-ante whether to denominate their short-term debt in euros or in dollars, this does not necessarily rule out the possibility of multiple equilibria.

Related Literature. This study relates to several broad strands of the literature. First, it is directly related to papers studying self-fulfilling crises in open economies, starting with [Calvo \(1988\)](#) and followed by [Obstfeld \(1996\)](#), [Cole and Kehoe \(2000\)](#), and more recently by [Céspedes et al. \(2017\)](#), [Aguilar et al. \(2017\)](#), [Farhi and Maggiori \(2018\)](#), [Fornaro \(2022\)](#), [Bianchi and Coulibaly \(2023\)](#) and [Bocola and Lorenzoni \(2020\)](#), among others. The feedback loop that drives the results in my framework works similarly as in a “third-generation” currency crisis model ([Krugman, 1999](#)), but with a few important differences. These types of models have mostly been used to study emerging markets, and were particularly relevant to understanding the financial crises that they faced during the 90’s. The novelty of this paper is that it focuses on global banks in a large economy such as the EU, which brings up two main differences with respect to traditional models. On the one hand, the liquidity shortages these banks face come from their maturity mismatches in dollars, rather than from currency mismatches, as in most emerging economies. Moreover, the collapse of these global intermediaries has significant spillovers to the international financial system, particularly to the US.

Given the role that the Fed plays in my model, this paper relates closely to the literature on bank-runs and the benefits of a lender of last resort, as in [Bagehot \(1873\)](#),

³This result aligns with the argument in [Farhi and Maggiori \(2018\)](#) regarding the possibility of the US underproviding safe assets as the dominant global issuer.

Diamond and Dybvig (1983), or Rochet and Vives (2004). In recent decades, there has been a growing attention towards the need of an international lender of last resort, as for example in Fischer (1999), Goodhart and Huang (2000), Mishkin (2001), Lerrick and Meltzer (2003), and more recently in Obstfeld (2009), Landau (2014), Cecchetti (2014), McDowell (2017), among others. I argue that the Fed is better equipped than any institution to fulfill this role, given that the majority of the liquidity needs of the international financial system are denominated in dollars. Moreover, contrary to traditional models that focus on runs on individual banks, I consider runs on the entire banking system that are linked to macroeconomic factors such as the exchange rate.

The focus on global banking of this paper is shared with a growing set of mostly empirical studies⁴ (Cetorelli and Goldberg, 2012; Shin, 2012; Bräuning and Ivashina, 2020; Aldasoro et al., 2019). The behavior of global banks and their role in the transmission of crises are modelled in Kalemli-Ozcan et al. (2013), Ivashina et al. (2015) and Morelli et al. (2022). I follow a similar theoretical approach to Gabaix and Maggiori (2015) in building a minimalistic real model with two countries, financial frictions, and global financial intermediaries at the centre of the capital flows and the exchange rate determination. However, they do not consider potential imbalances in the balance sheet of the intermediaries, which in my model open the door to multiple equilibria and benefits from an international lender of last resort.

In this context, a key feature of this paper when assessing the role of global banks is the dollar dominance. Recent studies that incorporate this characteristic, especially when focusing on exchange rate determination include Bruno and Shin (2015), Gourinchas et al. (2010), Maggiori (2017), Itskhoki and Mukhin (2021), Bianchi et al. (2021), Kekre and Lenel (2024), among others. Many of these studies focus on the US as the “banker to the world”, providing safe assets to the world. However, this traditional view predicts a dollar depreciation in times of crisis—contradicting empirical evidence and giving rise to the “reserve currency paradox”⁵—which the authors try to challenge by incorporating “flight-to-safety” shocks (Kekre and Lenel, 2024) or exogenous trade costs that are linked to banks’ health (Maggiori, 2017), to mention a few. In contrast, this paper shifts the focus to non-US global banks, which played a crucial role in the intermediation of capital flows across developed countries in the run-up to the GFC. By doing so, the model is able to jointly explain the dollars’ role as the reserve currency and its particular dynamics

⁴Others include Acharya and Schnabl (2010), Correa et al. (2016), McGuire and von Peter (2012).

⁵The “reserve currency paradox” refers to the tension between the US dollar’s role as a global safe asset and the direction of wealth transfers during crises. In periods of global stress, the dollar appreciates, leading to a wealth transfer from the United States to the rest of the world (RoW), which holds large quantities of dollar-denominated assets. However, standard international macro models predict that such a transfer should cause the dollar to depreciate, as wealthier RoW agents increase demand for their own goods, affecting the terms of trade. Moreover, if the dollar depreciated in bad times, it would behave like a risky asset—paying off poorly in crises—contradicting its role as a global safe store of value. See Maggiori (2017).

during a global crisis.

Finally, this paper also relates to the stream of literature on swap lines, most of which takes a micro-level empirical approach. From the studies focusing on this intervention⁶ during the GFC, such as [Baba and Packer \(2009a\)](#), [Baba and Packer \(2009b\)](#), [Moessner and Allen \(2013\)](#), and [Aizenman and Pasricha \(2010\)](#), perhaps the most comprehensive study so far is [Bahaj and Reis \(2022a\)](#), who rely on a difference-in-difference identification to assess the effect of the Fed’s swap lines on CIP deviations, portfolio flows, and the price of dollar-denominated corporate bonds. In a related article ([Bahaj and Reis, 2020](#)), they study the impact on funding costs of the new swap lines introduced by the Fed during the Covid-19 pandemic, similarly to [Aizenman et al. \(2021\)](#), [Goldberg and Ravazzolo \(2022\)](#), and [Ferrara et al. \(2022\)](#). Considering all these studies, the overall consensus points to the swap lines effectively helping to ease strains in US dollar funding markets and addressing sudden stop type episodes for banking systems.

On the theory side, the number of references is more limited. [Bahaj and Reis \(2022a\)](#) provide a model of the market for FX forwards into a small-scale general equilibrium model and find that the Fed swap lines reduce bank funding risk and increase the investment in dollar-denominated assets of non-US banks. [Cesa-Bianchi et al. \(2022\)](#) on the other hand, propose a medium-scale DSGE model with a bank currency portfolio problem to assess the capacity of the swap lines to mitigate the impact of dollar-shortage shocks to the economy and financial system. [Kekre and Lenel \(2024\)](#) find that, in a business cycle model of the international monetary system, “flight-to-safety” shocks generate a dollar appreciation and a decline in global output, and show that dollar swap lines help to mitigate these effects. Contrary to the others listed here⁷, my paper offers a tractable model of the global economy that features multiple equilibria. This allows me to study the intervention from the perspective of a lender of last resort and as an instrument to prevent self-fulfilling crises.

The rest of the paper is organized as follows. Section 2 presents stylized facts that serve as the basis of the model. Section 3 describes the baseline model and Section 4 discusses the multiple equilibria that might arise under this framework. The benefits of an international lender of last resort are presented in Section 5, while Section 6 extends the model and provides an analysis of the incentives that motivate the intervention. Finally, Section 7 concludes.

⁶An older literature studied the swap lines that supported the Bretton Woods system as well as the Fed’s reciprocal swap system between 1962 and 1998, when they were mainly used to finance foreign exchange rate interventions and keep currencies pegged to the dollar (e.g. [Williamson, 1983](#); [Obstfeld et al., 2009](#)).

⁷[Bacchetta et al. \(2025\)](#) provide a model to analyze the adjustment of non-US global banks to dollar funding shocks. The focus of this paper is on the policy implications, especially on the incentives behind the Fed providing dollar liquidity to non-US banks.

2 Stylized Facts

In this section I present three empirical facts that are distinctive features of the international financial system, and discuss briefly how I capture them in my model. Next, I elaborate on the usage and magnitude of the dollar swap lines, including the aspects of the intervention that I will highlight in the model.

Fact 1: Non-US global banks are key players in dollar markets.

Banks headquartered outside the United States hold trillions in dollar-denominated assets (Figure 1a), rivaling the size of the US banking sector itself (Shin, 2012). Many of these assets represent claims on US borrowers (\$7 trillion in 2021, as shown in Figure 1b), making non-US banks crucial intermediaries in dollar credit markets. I incorporate this fact in my model by having non-US global banks investing in productive long-term assets in the US.

Fact 2: Dollar funding of these banks is short-term and fragile.

The dollar assets of global banks are typically long-term and illiquid, but funded through short-term instruments. This maturity mismatch creates a structural vulnerability to liquidity shocks. In normal times, banks roll over their short-term liabilities through interbank markets, borrowing from money market funds (MMFs), or hedging in FX markets. But during stress episodes, these funding sources can dry up suddenly⁸. McGuire and von Peter (2012) document that European banks' short term dollar funding gap (i.e. dollar roll-over needs) was at least 7% of US GDP at the onset of the GFC.

Figure 2a shows the net dollar positions of non-US global banks by counterparty sector⁹. The persistent net lending position vis-à-vis non-banks highlights the scale of long-term investments in dollars. These positions, which have been consistently increasing since 2009, are offset by net borrowing from banks, derivatives markets, and monetary authorities—forms of funding that are inherently short-term and potentially unstable. Motivated by these characteristics, the model features global banks that are solvent in dollars, but have short-term dollar liabilities that need to be rolled-over.

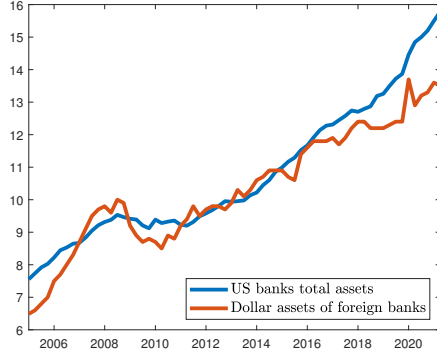
Fact 3: The dollar appreciates and liquidity dries up during a global crisis.

Periods of systemic stress, such as the GFC and the Covid-19 shock, are marked by rising funding costs and a sharp dollar appreciation (see Appendix A). While higher fund-

⁸Aldasoro et al. (2021) show that with around \$1.4 trillion, US and offshore money market funds represented around 12% of the on-balance sheet dollar funding for non-US banks at end-2019. MMFs are a flighty funding source: Figure 14 shows that non-US banks lost around \$300 billion during the covid-19 turmoil, mostly from US markets.

⁹Precisely calculating the short-term dollar needs of global banks is challenging due to the lack of data on banks' positions broken down simultaneously by nationality, currency, and maturity. Following the literature (McGuire et al., 2009, Aldasoro et al., 2020), however, a proxy measure of short-term dollar needs can be obtained using the BIS's counterparty sector classification. See Online Appendix A1 for a detailed explanation.

(a) Dollar cross-border foreign currency claims and US banks' total assets (\$ trillions)



(b) Foreign claims of BIS reporting banks on US counterparties (\$ trillions)

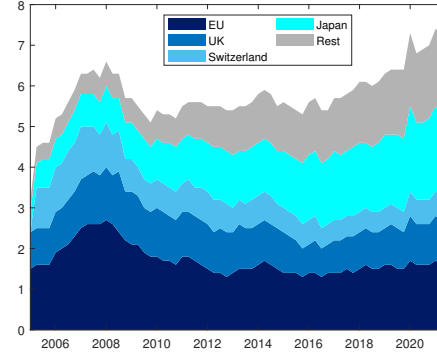
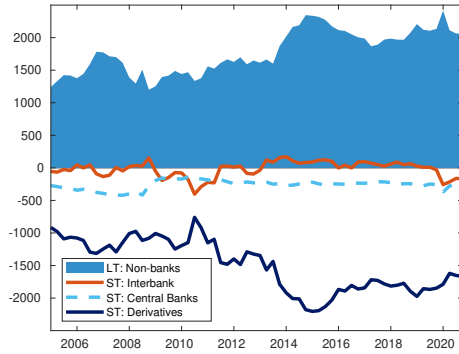


Figure 1: Dollar intermediation of non-US global banks

Note: Panel (a) considers US chartered commercial banks' total financial assets, and US dollar assets of banks outside the US. Panel (b) considers banks' nationality, i.e. the country where the headquarters are located, irrespective of the location of the reporting banks, and is based on the BIS Locational Statistics. **Source:** BIS, Flow of Funds, Fed.

(a) Net dollar-denominated positions of selected BIS reporting banks (\$ billions)



(b) Outstanding swap lines usage, by central bank (\$ billions)

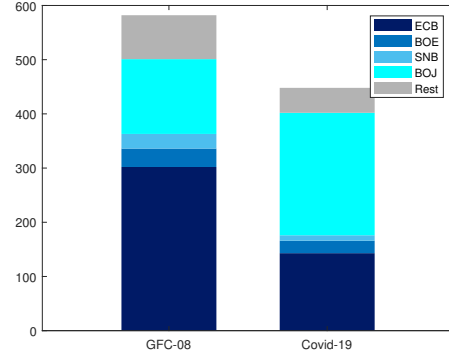


Figure 2: Dollar funding of non-US global banks

Note: Panel (a) considers cross-border and domestic positions in dollars of some of the countries with the largest banking sectors globally (see Online Appendix A1 for details). LT: long-term; ST: short-term; Interbank: net positions with other (unaffiliated) banks; Monetary Authorities: net cross-border liabilities to monetary authorities (mainly deposits of foreign exchange reserves); Non-banks: net positions vis-à-vis all non-banks; Derivatives: cross-currency positions, mostly FX swaps, calculated as the difference between total US dollar assets and liabilities. Panel (b) displays the peak usage of the Federal Reserve's swap lines during the global financial crisis (February 2009) and the COVID-19 pandemic (March 2020). **Source:** BIS, New York Fed, author's calculations.

ing costs are expected in crisis times, the persistent dollar strength is a well-documented fact that traditional macro-finance models fail to capture, known as the reserve currency paradox (Maggiore, 2017). I introduce this fact in the model through a financial friction that limits banks' ability to raise short-term funds. When banks face maturity mismatches in dollars, a stronger dollar increases the burden of their dollar liabilities more

than the pledgeable value of future asset returns, tightening their financial constraints.

Fact 4: Non-US global banks benefit from Fed liquidity in times of stress.

Since the GFC, the Federal Reserve has provided dollar liquidity to foreign central banks via swap lines, enabling them to support their domestic banks during episodes of global funding stress. Usage of these facilities peaked at over \$550 billion in 2009 and again at around \$450 billion in 2020 (see Figure 2b). The ECB and the Bank of Japan accounted for the largest drawdowns, reflecting the size of their banking systems' dollar exposures. The cross-country pattern of usage closely mirrors the scale of non-US banks' dollar claims on US borrowers, suggesting that the Fed's intervention may have been aimed not only at preventing fire sales of dollar-denominated assets, but also at averting a contraction in lending by foreign banks within the US. In the model I focus mostly on the last motive as the incentive for the Fed to provide dollar liquidity to the world.

3 Model

This section presents a two-period model of the world economy with financial frictions and cross-border intermediation. The global economy consists of two regions—the United States (US) and the Euro area (EU)—each populated by a continuum of households and endowed with both tradable and non-tradable goods. There are three goods in total: one tradable, exchanged internationally, and one non-tradable specific to each region. The non-tradable good serves as the numéraire within its respective economy. Following [Gabaix and Maggiori \(2015\)](#), I interpret the words *dollar*- and *euro*-denominated as values expressed in units of US and EU non-tradable goods, respectively. A continuum of global banks—owned by EU households—borrow in domestic currencies¹⁰ from both EU and US households and invest in long-term assets across countries.

The model is built around three key ingredients. First, global banks facilitate the cross-border financial transactions resulting from households' saving decisions. Second, they face an agency friction that limits their ability to raise funds and to roll-over their debt in order to continue their operations. Third, their portfolio consists on short-term liabilities and illiquid long-term assets such that a maturity mismatch in dollars is formed. The last two ingredients combined result in tighter financial conditions if the dollar appreciates in the short-run, and may even trigger a system-wide bank-run. Also, because global banks invest in long-term assets in the US, their operations have spillovers towards the US economy by boosting future non-tradable output.

The remainder of this section formalizes the agents' behavior and discusses some

¹⁰This is in line with the empirical evidence provided -for example- by [Maggiori et al. \(2020\)](#), in which they establish that investor holdings are biased toward their own currencies to such an extent that countries typically hold most of the foreign debt securities denominated in their currency.

simplifying assumptions.

3.1 Households

Euro area households derive utility from consuming tradable and non-tradable goods, C_t^T and C_t^N , which are combined in a consumption basket defined as $C_t \equiv (C_t^N)^{1-\omega}(C_t^T)^\omega$. The parameter $0 < \omega < 1$ denotes their preference for the tradable good, which has a relative price of p_t with respect to the non-tradable good in the EU. Goods markets for tradables are frictionless across countries, but non-tradables are traded only domestically.

Financial markets are incomplete. Households can save either in euro-denominated bonds B issued by global banks, or in dollar-denominated bonds $\tilde{B}$ traded directly with US households. I assume a simplified portfolio structure: households choose only one form of saving depending on whether banks are operating. When banks are active, households invest via B ; otherwise, they rely on direct dollar bond holdings $\tilde{B}$.¹¹ Since EU households own the banks, they receive bank profits Π in period 2.

The household's problem is:

$$\max_{C_t} U = \ln(C_1) + \beta \mathbb{E} \ln(C_2)$$

subject to the budget constraints:

$$Y_1^N + p_1 Y_1^T + \mathcal{L} = C_1^N + p_1 C_1^T + \mathcal{B} \quad (1)$$

$$\Pi + Y_2^N + p_2 Y_2^T + \mathcal{R}\mathcal{B} = C_2^N + p_2 C_2^T \quad (2)$$

where Y_t^T and Y_t^N are the households' endowments of the tradable and non-tradable goods, respectively. The euro value of purchases is $\mathcal{B} \in \{B, e_1 \tilde{B}\}$, and their corresponding repayment is $\mathcal{R}\mathcal{B} \in \{RB, R^* e_2 \tilde{B}\}$, where R and R^* are the gross returns on euro- and dollar-denominated bonds, and e_t is the exchange rate. The term $\mathcal{L}$ is the payment received from a pre-existing position with global banks, whose value depends on whether a run on the banking system occurs.

A key variable in this real model is the exchange rate e_t , defined as the relative price between the two non-tradable goods, or in other words, as the quantity of *euros* bought by one *dollar*. Consequently, an increase in e_t represents a dollar appreciation.

For ease of exposition, I begin with a baseline in which all bond holdings are intermediated by global banks (i.e., $\mathcal{B} = B$). The resulting optimality conditions are:

¹¹This can be rationalized by assuming a small non-pecuniary cost associated with direct cross-border bond trading. For instance, households may lack the expertise and financial sophistication of global banks, which creates frictions in international financial access, as discussed in [Brunnermeier and Sannikov \(2014\)](#). A formal version of this problem is presented in Online Appendix [A2](#).

$$p_1 C_1^T = \frac{1}{\beta R} p_2 C_2^T \quad (3)$$

$$p_t = \frac{C_t^N}{C_t^T} \frac{\omega}{1 - \omega} , \quad (4)$$

where (3) is the Euler equation in terms of tradable consumption and prices, and (4) determines the optimal allocation of consumption expenditure between tradable and non-tradable goods.

US households face an analogous optimization problem. They derive utility from consumption according to $U^* = \ln(C_1^*) + \beta \ln(C_2^*)$, where starred variables denote US quantities. Their basket of consumption is given by $C_t^* \equiv (C_t^{*N})^{1-\omega^*} (C_t^{*T})^{\omega^*}$, and they also receive endowments of both goods. Their utility maximization problem is reported in the Online Appendix A2.

The key differences relative to EU households are that US households: (i) save in bonds denominated in dollars—either with global banks (B^*) or directly with EU households ($\tilde{B}^*$), (ii) hold pre-existing dollar claims $\mathcal{L}^*$ on global banks, and (iii) do not receive bank profits. Their first-order conditions follow the same intuition as their EU counterpart, and are given by

$$p_1^* C_1^{*T} = \frac{1}{\beta^* R^*} p_2^* C_2^{*T} \quad (5)$$

$$p_t^* = \frac{C_t^{*N}}{C_t^{*T}} \frac{\omega^*}{1 - \omega^*} . \quad (6)$$

An important feature of the model is that the value of households' pre-existing positions with global banks relies on the banks' ability to fulfill their repayment obligations. Under normal conditions, these positions are fully repaid. However, in the event of an unanticipated liquidation, only a fraction is recovered. Formally, we can express this as

$$\mathcal{L} = \begin{cases} L & \text{if banks are not liquidated,} \\ \Omega L & \text{if banks are liquidated,} \end{cases} \quad (7)$$

and analogously for $\mathcal{L}^*$, with $L, L^* \geq 0$, and with $\Omega, \Omega^* \in [0, 1]$ denoting the recovered fractions.

3.2 Global Banks

Global banks are owned by EU households¹² and serve two primary functions: they intermediate cross-border financial flows and invest in long-term projects that expand the supply of non-tradable goods in both economies. Rather than modeling banks' funding and investment decisions explicitly¹³, I assume they begin with pre-existing balance sheet positions, consistent with the empirical facts discussed in Section 2.

Banks enter period 1 with short-term liabilities denominated in euros and dollars, L and L^* , which must be repaid to households. To do so, they must raise funds by issuing new euro- and dollar-denominated bonds, B and B^* , to EU and US households, respectively. A necessary condition for continued operation is then

$$L + e_1 L^* \leq B + e_1 B^* . \quad (8)$$

Banks' assets are long-term investments¹⁴ with gross returns $\mathcal{A}_t$ and $\mathcal{A}_t^*$ in euros and in dollars, respectively. Under normal circumstances, they are held until maturity in period 2, but their gross returns depend on whether banks are liquidated or not, as it will become clear later. Banks' profits in period 2 are then given by

$$\Pi = \mathcal{A}_2 + e_2 \mathcal{A}_2^* - RB - e_2 R^* B^* . \quad (9)$$

The last two equations are expressed in euros, which is why dollar quantities are multiplied by the corresponding exchange rate. Finally, banks face an agency friction that limits their ability to raise funds and might prevent them from rolling-over their initial debt. In each period, after taking positions, they can divert a fraction of the funds they intermediate. If they divert the funds, banks are unwound and the households that had lent to them in $t = 1$ recover a portion $1 - \gamma \geq 0$ of their credit position $B + e_1 B^*$. Since creditors—when lending to the banks—correctly anticipate their incentives to divert funds, banks are subject to a credit constraint of the form

$$\frac{1}{R} \Pi \geq \gamma (B + e_1 B^*) , \quad (10)$$

where $1/R$ comes from EU households' stochastic discount factor. Since the investment is fixed, bankers simply choose a combination of B and B^* to maximize the expected

¹²The reader can think of a US banking sector also operating in the background. In a more complex set up, these banks would intermediate the funds from US households, and then engage in cross-border operations with foreign banks. Since the model focuses on non-US global banks and their balance sheet mismatches, the lending by US banks is immaterial to the results, so I leave it unspecified.

¹³Online Appendix A6 extends the model to endogenize these decisions.

¹⁴Since they are denominated in non-tradable goods, these assets can be thought as an investment in the housing sector. They can also be interpreted as if banks were financing firms that invest in the non-tradable sector.

profits in (9) subject to the liquidity needs in (8) and the financial constraint in (10). The optimization problem results in the following no-arbitrage condition:

$$R = R^* \frac{e_2}{e_1} \quad (11)$$

which reflects that the uncovered interest parity (UIP) holds¹⁵.

Costly Liquidation and Repayment

If households expect the credit constraint of banks to be violated, they do not provide deposits in period 1 (i.e., $B = B^* = 0$), preventing the liquidity needs in (8) from being met. As a result, global banks are forced to shut down and liquidate their long-term assets prematurely.

These assets are illiquid and lose value when liquidated. Let $\mathcal{A}_1, \mathcal{A}_2$ and $\mathcal{A}_1^*, \mathcal{A}_2^*$ denote the state-contingent euro- and dollar-denominated asset values at periods 1 and 2. Their values are:

$$(\mathcal{A}_1, \mathcal{A}_2) = \begin{cases} (0, A) & \text{if banks operate} \\ (\Psi A, 0) & \text{if banks shut down} \end{cases} \quad (12)$$

and analogously for $\mathcal{A}_1^*, \mathcal{A}_2^*$, with $\Psi, \Psi^* \in [0, 1]$ denoting the fraction of the long-term investment value that can be recovered through fire sales.

If forced to shut down, banks recover only $\Psi A + e_1 \Psi^* A^*$ in period 1. These proceeds are fully used to repay pre-existing liabilities, with repayment rates Ω, Ω^* that satisfy

$$\Psi A + e_1 \Psi^* A^* = \Omega L + e_1 \Omega^* L^*. \quad (13)$$

Without loss of generality¹⁶ I treat $\Omega^* \in [0, 1]$ as a parameter of the model, while letting Ω adjust to satisfy the repayment condition in (13). Together with (9), this implies that banks make no profits in any period if they shut down, $\Pi = 0$. This structure captures the limited recovery associated with forced liquidation and the tight link between asset fire sales and household losses, also present, for instance, in [Ivashina et al. \(2015\)](#).

Exchange rate and banks' soundness

The exchange rate plays a key role in determining the soundness of global banks. A scenario in which all banks operate must necessarily satisfy equation (8) with equality¹⁷.

¹⁵This no arbitrage condition arises from the fact that banks take R and R^* as given.

¹⁶This choice avoids pinning down the exchange rate mechanically, and is without loss of generality: the reverse treatment—fixing Ω and solving for Ω^* would yield similar results.

¹⁷In $t = 1$, banks only intermediate flows across countries, and do not invest. Therefore, in equilibrium, gross capital flows in both countries have to offset each other.

Combining the two restrictions, the UIP condition, and banks' profits yields the following expression for the financial constraint, in terms of e_1 :

$$\frac{A}{R} + e_1 \frac{A^*}{R^*} \geq (1 + \gamma)(L + e_1 L^*) .$$

The magnitude and direction of the impact of exchange rate fluctuations on the incentive compatibility constraint depends on the composition of banks' balance sheet. I will follow the literature of bank runs, where financial intermediaries might face liquidity issues but are otherwise solvent. Particularly, I will focus on the case where banks are solvent in dollars, but exposed to dollar liquidity shortages. This can be captured in the model by making the following assumptions.

Assumption 1. *The following conditions on the dollar portfolio of global banks hold:*

$$\begin{aligned} \text{Dollar profitability:} \quad & \frac{A^*}{R^*} - L^* > 0 \\ \text{Dollar liquidity:} \quad & \frac{A^*}{R^*} - (1 + \gamma)L^* < 0 \end{aligned}$$

The previous two inequalities reflect that the discounted dollar profits of global banks might be large compared to their current dollar liabilities, suggesting no currency mismatches. However, they might be insufficient to cover their short-term dollar needs, which are determined also by γ . It is possible to interpret this parameter as capturing the market's risk intolerance, so that liquidity needs are larger when this intolerance is higher. With these conditions, the incentive compatibility constraint in (10) leads to the following Lemma.

Lemma 3.1. *Suppose that Assumption 1 holds. A necessary condition for all banks to operate in equilibrium is,*

$$e_1 \leq \frac{A/R - (1 + \gamma)L}{(1 + \gamma)L^* - A^*/R^*} \equiv \bar{e} ,$$

where $R = (A + Y_2^N)/\beta Y_1^N$ and $R^* = (A^* + Y_2^{*N})/\beta^* Y_1^{*N}$ are the interest rates when banks operate.

Proof. In Appendix B.3. □

The threshold $\bar{e}$ can be interpreted as the maximum exchange rate that the banking system can tolerate¹⁸. This shows that, although e_1 affects the return of dollar investments positively, it also increases the burden of the dollar liabilities that banks need to roll-over, while the agency friction prevents them from pledging all of their future

¹⁸Even if we consider banks that start period 1 with assets denominated in tradable goods, they can still be exposed to dollar fluctuations, as shown in Online Appendix A4.1.

asset returns to secure refinancing¹⁹. Under Assumption 1, the overall result is that market conditions become tighter the higher is the exchange rate, in line with the evidence presented in Section 2. If the depreciation goes beyond the threshold $\bar{e}$, banks cannot roll-over their debt and go bust. Therefore, a scenario in which all global banks operate can only happen when $e_1 \leq \bar{e}$. In the next section I will discuss in detail the different equilibria that might arise in the model depending $\bar{e}$ and households' expectations of the exchange rate.

3.3 Market Clearing

Market clearing for the non-tradable consumption good requires:

$$Y_t^N + \mathcal{A}_t = C_t^N \quad Y_t^{*N} + \mathcal{A}_t^* = C_t^{*N}. \quad (14)$$

These equations capture the idea that the outcome of the long-term investments contributes to non-tradable output in both countries—either at $t = 2$ when assets mature as expected, or at $t = 1$ in the event of premature liquidation. In this sense, the assets can be interpreted as productive investment projects that augment the supply of non-tradable goods.

The market clearing condition for the tradable good requires that the world's endowment is equal to the world's demand in both periods,

$$Y_t^T + Y_t^{*T} = C_t^T + C_t^{*T}. \quad (15)$$

The share of EU endowment of the tradable good in the world economy is denoted as

$$\eta_t \equiv \frac{Y_t^T}{Y_t^T + Y_t^{*T}},$$

and equivalently for the US, such that $1 - \eta_t = \eta_t^*$. I assume that all prices are flexible and that the law of one price holds, such that

$$e_t p_t^* = p_t. \quad (16)$$

Finally, market clearing in the international bond traded directly between EU and US households requires

$$\tilde{B} = -\tilde{B}^*. \quad (17)$$

¹⁹From this perspective, the financial constraint can be interpreted as preventing banks from pledging more than a fraction $\frac{1}{1+\gamma} < 1$ of the present value of their asset returns.

3.4 Simplifying assumptions

To streamline the algebra and focus on the key economic mechanisms, I assume symmetric preferences and discount factors across countries, such that $\omega = \omega^*$ and $\beta = \beta^*$. I also normalize non-tradable endowments in period 1 to $Y_1^N = Y_1^{*N} = 1$.

On the financial side, to narrow the analysis to dollar shortages, I will assume for now that there is no euro-denominated debt that has to be rolled-over, and that the liquidation process is fully unproductive in the event of a bank-run such that no repayment is made²⁰. Formally,

$$L = 0, \quad \Psi = \Psi^* = 0, \quad \Omega^* = 0.$$

Overall, in this baseline specification, a run leads to a complete loss of assets and no repayment to creditors. In Section 6 I provide a generalization of the model that relaxes these assumptions, maintaining the main results but highlighting the general equilibrium effects at play.

4 Multiple equilibria and self-fulfilling crises

A key feature of the model is that global banks operate only if households expect their incentive compatibility constraint (10) to be satisfied. These expectations, and thus households' funding decisions, are shaped by their beliefs about the exchange rate at the beginning of period 1.

Exchange rate as a coordination device. Unlike standard bank-run models—where depositors coordinate over the actions of others to decide whether to withdraw or roll over their funds—this framework focuses on expectations about an aggregate variable that determines the viability of financial intermediation. In particular, households use the expected exchange rate as a coordination device. This mechanism is in line with empirical evidence showing that the strength of the US dollar acts as a barometer of global dollar credit conditions, with appreciations typically associated with tighter funding constraints (Bruno and Shin, 2023).

Two distinct equilibria may arise from this coordination problem.

4.1 No-run equilibrium

In a *no-run equilibrium*, households expect a sufficiently weak dollar—specifically, $e_1 < \bar{e}$ as identified in Lemma 3.1—such that the bank incentive compatibility constraint (10) is satisfied. Anticipating that banks will not divert funds, households choose to provide financing in period 1. As a result, global banks operate, roll over their liabilities, and allow

²⁰Since $L = 0$, the value of Ω is immaterial in this baseline specification.

investments to mature in period 2. Liabilities are repaid in full, and profits are distributed to EU households. This scenario is similar to the no-run equilibrium in classic bank-run models, which definition is the following.

Definition 1 (Competitive no-run Equilibrium). *A competitive no-run equilibrium is a path of real allocations $\{C_t^T, C_t^N, C_t^{*T}, C_t^{*N}\}_t$, bonds $\{B, B^*\}$, bank variables $\{\Pi, \mathcal{A}_t, \mathcal{A}_t^*, \mathcal{L}, \mathcal{L}^*\}_t$, and prices $\{R, R^*, e_t, p_t, p_t^*\}_t$, satisfying the households' optimality conditions in (1), (2), (3) and (4) -plus their counterparts for the US economy-, the banks' roll-over needs, profits, credit constraint, and no-arbitrage condition in (8),(9),(10) and (11), the market clearing conditions in (14) and (15), and the assets and liabilities determination in (7) and (12) -plus their counterparts for the US- given a path of endowments $\{Y_t^T, Y_t^N, Y_t^{*T}, Y_t^{*N}\}_t$.*

I will refer to the “no-run” equilibrium as the “good” equilibrium, with an exchange rate in $t = 1$ denoted by e_1^G .

4.2 Run equilibrium

Contrary to the previous case, it is optimal for households not to save with global banks if they expect condition (10) to be violated. This is the case if they expect a relatively strong dollar in $t = 1$ ($e_1 > \bar{e}$) that would increase the incentives of banks to divert their funds, as explained in Section 3. Under these circumstances, banks collapse, their investment in US and EU assets is lost, and their profits Π become null. Importantly, in the absence of banks, households resort to trading dollar bonds directly across borders. Therefore, the Euler equation for EU households in (3) that considered euro-denominated bonds, is replaced by

$$p_1 C_1^T = \frac{1}{\beta R^*} \cdot \frac{e_1}{e_2} p_2 C_2^T . \quad (18)$$

This equilibrium is defined as follows.

Definition 2 (Competitive run Equilibrium). *A competitive run equilibrium is a path of real allocations $\{C_t^T, C_t^N, C_t^{*T}, C_t^{*N}\}_t$, bonds $\{\tilde{B}, \tilde{B}^*\}$, bank variables $\{\Pi, \mathcal{A}_t, \mathcal{A}_t^*, \mathcal{L}, \mathcal{L}^*, \Omega\}_t$, and prices $\{R^*, e_t, p_t, p_t^*\}_t$, satisfying the households' optimality conditions in (1), (2), (4) and (18) -plus their counterparts for the US economy-, the banks' profits and repayment in (9) and (13), the market clearing conditions in (14) and (15), and the assets and liabilities determination in (7) and (12) -plus their counterparts for the US- given a path of endowments $\{Y_t^T, Y_t^N, Y_t^{*T}, Y_t^{*N}\}_t$.*

I will refer to the “run” equilibrium as the “bad” equilibrium, with an exchange rate in $t = 1$ denoted by e_1^B .

4.3 Multiple Equilibria

To give a better sense of the forces driving the equilibria of the model, I will fully characterize them using two variables, the exchange rate and capital flows, and two equations. Both variables play a key role in financial crises, and eventually will drive most of intuition behind the main results of the model. In particular, I will focus on the exchange rate in period 1, and on EU savings/debt, $\mathcal{B}$. Since this is a two-country model, EU net savings are equivalent to capital flows to the US, so I will use both terms interchangeably.

4.3.1 Static determination of exchange rates and capital flows

First, I will analyze how capital flows affect the exchange rate in period 1. From the perspective of the EU, the trade balance -in euros- is defined as follows:

$$p_1(Y_1^T - C_1^T) = \mathcal{B} ,$$

where $\mathcal{B}$ are the net capital flows to the US. Focusing on the left-hand side of the previous expression, the households' optimality condition in (4) and the market clearing conditions for non-tradable goods tell us that their expenditure in tradables is fixed, so that $p_1 C_1^T = \frac{\omega}{1-\omega} Y_1^N = \frac{\omega}{1-\omega}$. Furthermore, simple derivations presented in the appendix show that tradable market clearing (15) and utility maximization imply that

$$p_1 = \frac{\omega}{1-\omega} \frac{1}{Y_1^T + Y_1^{*T}} (1 + e_1) ,$$

reflecting the fact that, when a country's exchange rate depreciates, consuming tradable goods becomes more expensive. Finally, rearranging the equations above to express e_1 as a function of $\mathcal{B}$ yields

$$e_1(\mathcal{B}) = \underbrace{\frac{\eta_1^*}{\eta_1}}_{\text{Endowment component}} + \underbrace{\mathcal{B} \cdot \frac{1-\omega}{\omega} \cdot \frac{1}{\eta_1}}_{\text{Capital flows component}} . \quad (19)$$

This equation describes a very intuitive result. The first component shows that, absent capital flows, the exchange rate is determined simply by the relative endowment of tradable goods in each economy. More interestingly, the second component captures the idea that the larger the capital outflows towards the US (EU savings), the larger the trade balance that the EU needs in period 1 to cover those outflows. Ultimately, a stronger trade balance is achieved by a euro depreciation ($\uparrow e_1$). Another way to look at this idea is that a weaker euro makes EU exports more attractive in markets abroad.

4.3.2 Intertemporal determination of exchange rates and capital flows

Now let us consider how the exchange rate in period 2 affects capital flows. When banks operate, EU households receive their profits and thus the budget constraint they face in period 2 is

$$R \cdot B = p_2(C_2^T - Y_2^T) + C_2^N - Y_2^N - \Pi .$$

Following a similar procedure as for the trade balance in period 1, it is possible to rewrite their expenditure in tradables as $p_2 C_2^T = \frac{\omega}{1-\omega} C_2^N$ and the price of tradables as

$$p_2 = \frac{\omega}{1-\omega} \frac{1}{Y_2^T + Y_2^{*T}} (C_2^N + e_2 C_2^{*N}).$$

Simple derivations described in the appendix show that the previous equation can be written in terms of e_1 by using the expressions for both interest rates, the UIP condition $e_2 R^* = e_1 R$, and the market clearing conditions for non-tradable goods, $C_2^N = Y_2^N + A$ and $C_2^{*N} = Y_2^{*N} + A^*$. Next, banks' profits Π given by equation (9) can also be expressed in terms of e_1 by using the UIP condition and the roll-over needs in (8), so that $\Pi = R \left[e_1 \left(\frac{A^*}{R^*} - L^* \right) + \frac{A}{R} \right]$. Finally, combining all these expressions, the budget constraint in period 2 yields the following equation,

$$B(e_1) = \frac{\omega}{1-\omega} \beta \underbrace{\left(\eta_2^* - e_1 \eta_2 \right)}_{\text{Endowment component}} - e_1 \underbrace{\left(\frac{A^*}{R^*} - L^* \right)}_{\text{Dollar profits}} .$$

The previous equation highlights the importance of wealth effects in determining capital flows and exchange rates. The first term on the right-hand side shows that an exchange rate appreciation ($\downarrow e_2$) in the EU represents a drop in relative prices in that economy in period 2, which pushes EU households to increase future consumption by saving more (or borrowing less) in period 1, thus increasing capital outflows. This can be thought of in terms of e_1 . In anticipation of the drop in prices and thus higher relative wealth, households will increase consumption today as well, which pushes the euro to also appreciate in period 1. The last term on the right-hand side shows that a dollar appreciation in $t = 2$ represents a positive wealth shock for EU households if the dollar profits they receive from banks are positive, as by Assumption 1. This positive wealth effect reinforces the mechanism just discussed and leads to fewer capital outflows in $t = 1$, as EU households require less savings.

Now let us consider the scenario where global banks do not operate, which happens when $e_1 > \bar{e}$. In this context, there is one distinct force at play that will change the intertemporal relation between the exchange rate and capital flows. When banks go

bust, their profits collapse to $\Pi = 0$ because of the costly liquidation of their long-term assets, and the failure to repay their short-term liabilities, as discussed in Section 3. This represents a negative wealth effect for EU households in period 2, leading them to demand more savings (fewer capital outflows) and consume less in period 1, for a given level of exchange rate. Considering these two cases, and the fact that households use the exchange rate as a coordination device, the intertemporal relation between the exchange rate and capital flows can be characterized as follows:

$$\mathcal{B}(e_1) = \begin{cases} \frac{\omega}{1-\omega}\beta(\eta_2^* - e_1\eta_2) - e_1\left(\frac{A^*}{R^*} - L^*\right) & \text{if } e_1 < \bar{e} \\ \frac{\omega}{1-\omega}\beta(\eta_2^* - e_1\eta_2) & \text{if } e_1 > \bar{e} \end{cases} \quad (20)$$

The equilibria of the model can be obtained by solving the system of two equations given by (19) and (20). Using the properties of these two schedules, we can conclude the following.

Proposition 1. *Suppose that Assumption 1 holds, $\eta_t = \eta \forall t$, and let $\bar{e}$ be the value of e_1 that makes condition (10) hold with equality. Then, multiple equilibria are possible if*

$$\underbrace{\frac{\eta^*}{\eta + \frac{1}{1+\beta}\frac{1-\omega}{\omega}(A^*/R^* - L^*)}}_{e^G} \leq \underbrace{\frac{A/R}{(1+\gamma)L^* - A^*/R^*}}_{\bar{e}} \leq \underbrace{\frac{\eta^*}{\eta}}_{e^B}$$

where $R = (A + Y_2^N)/\beta Y_1^N$ and $R^* = (A^* + Y_2^{*N})/\beta^* Y_1^{*N}$.

Proof. In Appendix B.4. □

In Proposition 1, e^G and e^B represent the equilibrium exchange rate when banks operate and when they shut down, respectively. One important point to highlight is that banks' dollar profits make the equilibrium exchange rate lower, which shows that relative wealth matters to determine the strength of a country's currency. In this framework, if a country is relatively wealthier, its currency will appreciate.

Figure 3 provides one example of the schedules derived previously. As explained before, $e(B)$ is increasing in B from a trade balance perspective. An increase in capital outflows towards the US has to be compensated by a stronger trade balance in the EU, which is achieved by a euro depreciation. On the other hand, $\mathcal{B}(e_1)$ is decreasing in e_1 . However, if the dollar appreciates beyond $\bar{e}$, market conditions tighten to the point where banks shut down, affecting the EU economy and generating an abrupt contraction in capital flows. In cases in which this negative wealth effect is strong, multiple equilibria can arise. I interpret the “bad” equilibrium with a strong dollar (e^B) and collapsed banks as a financial crisis, and obtain a number of predictions about the behavior of

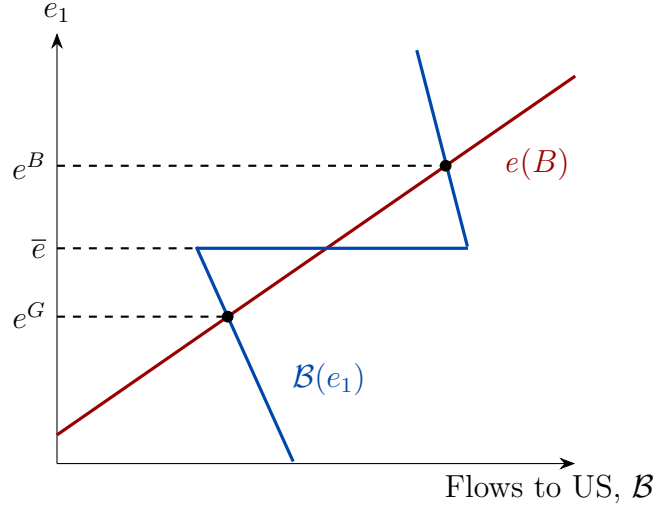


Figure 3: Multiple Equilibria

consumption, output, the exchange rate, and capital flows during those events. The next proposition collects these predictions.

Proposition 2. *If there are three equilibria and we compare the two stable ones, we obtain the following predictions about the crisis equilibrium with respect to the standard equilibrium:*

- i. The dollar is more appreciated;*
- ii. Banks face tighter financial conditions and struggle to roll over their debt;*
- iii. Global output and relative wealth in the EU are lower;*
- iv. Net capital flows to the US are larger.*

Proof. In Appendix B.5. □

These results are in line with the evidence provided in Section 2 and with other studies that rely on more complex models such as [Kekre and Lenel \(2024\)](#), [Cesa-Bianchi et al. \(2022\)](#) and [Maggiori \(2017\)](#). A crucial element in this mechanism is that the exchange rate depreciates when global banks collapse. In the model, this happens because global banks suffer a “sudden stop” during a crisis, which ends up hurting the aggregate demand in the EU and eventually depreciating the euro. In that sense, capital flows to the US increase ($\uparrow \mathcal{B}$), meaning that EU households have a higher willingness to save in $t = 1$, in anticipation of the reduction in wealth in the next period.

Self-fulfilling crises. In this context, expectations about e_1 —and the incentives of banks to divert funds—can become self-fulfilling. If households are pessimistic and expect a strong dollar, they will not provide banks with the funding to roll-over their debt, leading to a banking crisis in the EU and the loss of banks’ profits. Given the negative impact

on their relative wealth in period 2, EU households cut down consumption in $t = 1$ and increase savings, leading to a euro depreciation, confirming the initial expectations of a high exchange rate. Overall, this mechanism works because agents are atomistic and ignore the consequences that their actions have on aggregate outcomes²¹, as it is common in the literature studying self-fulfilling crises.

Importance of fundamentals

Notice that the existence of multiple equilibria depends on the fundamentals of the global economy. For example, when agents are impatient (low β), banks are more likely to divert funds, so that the dollar appreciation that makes banks collapse is even lower. Likewise, if the initial dollar short-term liabilities (L^*) are high, banks are more exposed to fluctuations in the exchange rate. Financial conditions also play a role: if they are tighter (high γ), the impact of an exchange rate depreciation on banks soundness is amplified, making multiple equilibria more possible²². To illustrate this, Figure 4 shows two cases when the model features only a unique equilibrium. In panel (a), the “good” equilibrium is the only one possible. On the contrary, in panel (b) only the “bad” equilibrium can materialize. Such a situation is likely if, for example, γ is particularly high and thus $\bar{e}$ shrinks, making global banks less resilient to exchange rate depreciations.

For completeness, panel (a) in Figure 5 shows how different values of γ give rise to the three potential scenarios for the economy. Recall that this parameter can be interpreted as the risk aversion of investors, thus $\bar{e}$ is decreasing in γ , but the values of the exchange rate in equilibrium are unchanged (e^G and e^B). The interesting case that this paper focuses on is one in which $\gamma' < \gamma < \gamma''$ so that the correspondence $\mathcal{C}^e$, which captures the potential values of e_1 in equilibrium, accepts both e^G and e^B as solutions. Panel (b) on the other hand, highlights the role of A^* and L^* on determining the equilibrium. A drop in A^* or an increase in L^* have similar effects: everything else constant, they lower $\bar{e}$ because of the increase in the dollar liquidity needs, and in addition, they increase e^G because of the lower profits of global banks and thus weaker demand from EU households. As a result, γ' and γ'' drop, enlarging the zone in which only the “bad” equilibrium materializes²³.

²¹The importance of lenders’ expectations for global banks is also highlighted in [Ivashina et al. \(2015\)](#), where they can have a significant impact on foreign banks that depend on unsecured short-term dollar funding, in the presence of frictions in the FX forward markets.

²²Online Appendix A3 presents a nominal version of the model to show that a monetary policy contraction in the US would also tighten banks’ financial constraint.

²³Even though in this simple framework I take the assets and liabilities of banks are given, the main results of the paper are unchanged if we endogenize their portfolio decision. Online Appendix A6 extends the basic model to show that even if non-US global banks could choose ex-ante whether to denominate their debt in euros or in dollars, this does not rule out the possibility of multiple equilibria. In other words, despite a maturity mismatch in dollars opens the door to a “bad” equilibrium, banks do not necessarily have sufficient ex-ante incentives to reduce their exchange rate exposure.

Dollar appreciation and asymmetric impact of global crises

The endogenous appreciation of the dollar in the model shows the importance of general equilibrium effects in shaping the asymmetric impact of global crises across countries. In normal times, a higher exchange rate e_1 increases global banks' profits from their net dollar position, $A^*/R^* - L^*$, effectively raising the relative wealth of EU households. This outcome is consistent with standard open economy models, which predict that a stronger dollar leads to a wealth transfer from the US to the rest of the world.

One of the model's key insight, however, lies in the reversal of this mechanism during crises²⁴. When shocks—such as a drop in asset values or a rise in global risk aversion—affect banks globally, institutions with short-term liabilities in a foreign currency are more vulnerable. If that currency appreciates, their funding burdens increase, potentially tightening their constraints and amplifying the original shock. Therefore, due to the dollar's dominant role in international funding markets, non-US banks are particularly exposed. In the model, this mechanism reduces the relative wealth of the EU, and reinforces the appreciation of the dollar—despite the initial shock being global in nature.

An additional factor contributing to this asymmetry is that, unlike US banks, non-US banks often lack a stable source of dollar funding—such as dollar-denominated deposits or direct access to liquidity facilities from their domestic central banks. This second issue is the main focus of Section 5.

4.4 Numerical example

I present a numerical example to illustrate how the model operates in a world exposed to multiple equilibria. The goal is to show how key variables respond to different economic conditions in equilibrium. Most parameters are calibrated to reflect the euro depreciation during the GFC, the dollar liquidity shortages faced by banks, and pre-crisis interest rates. Given the model's simplicity, this exercise should be viewed as illustrative rather than quantitative estimates.

One period corresponds to one quarter. The period I am particularly interested in modeling is Q4-2008, because this is when the US economy suffered its sharpest quarterly output decline since the late 50's, but the dollar rallied against most currencies, including the euro. Data for the US are retrieved from the U.S. Bureau of Economic Analysis and the Board of Governors of the Federal Reserve System, while data for the EU area comes from Eurostat. BIS is the source for the data on global banks.

The target pre-crisis annualized interest rates in the US and in the EU are 2.5% and 3.5%, respectively. This is meant to capture the low interest rate environment character-

²⁴Unlike much of the literature, this framework abstracts from the role of the dollar as a “safe haven”, since there is no uncertainty and agents have perfect foresight. Instead, the model highlights a distinct channel through which dollar appreciation affects the distribution of global wealth.

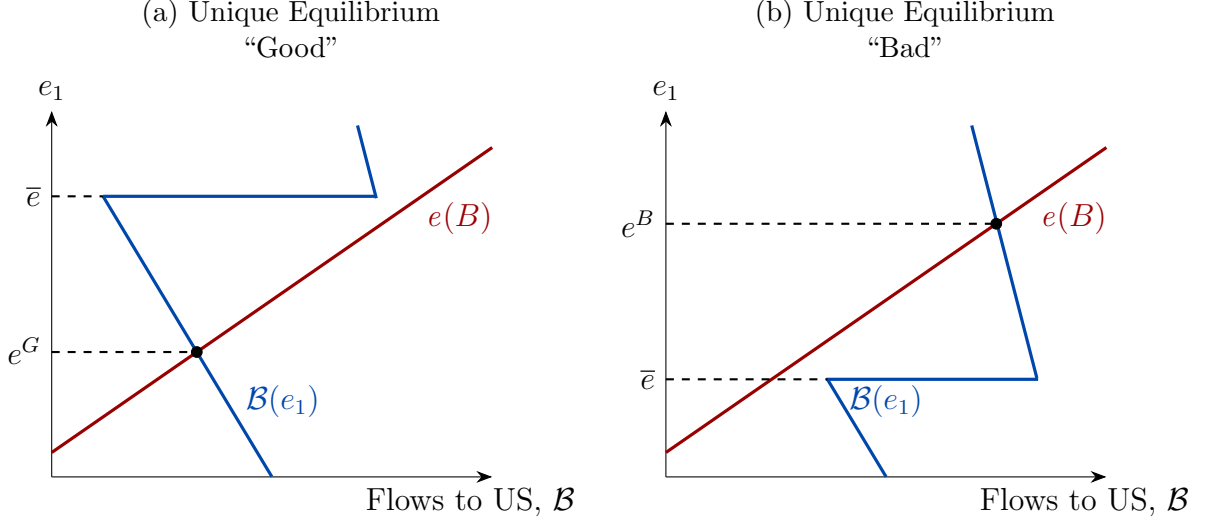


Figure 4: Unique Equilibria

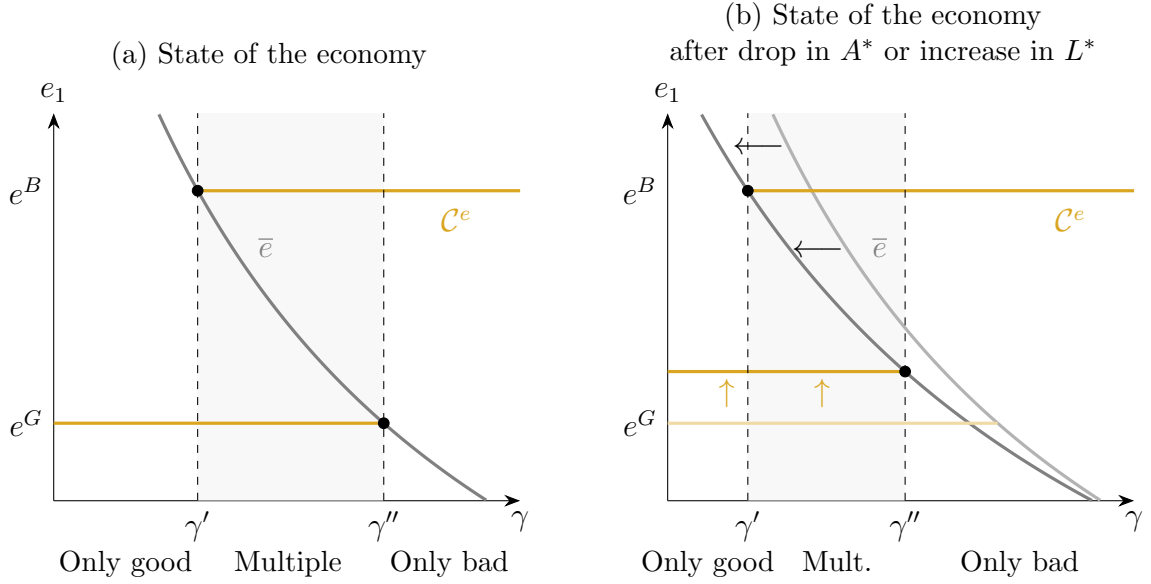


Figure 5: Exchange rate and severity of the financial friction

izing the world economy in the years preceding the start of the GFC. On the other hand, [McGuire and von Peter \(2012\)](#) estimate that the major European banks’ dollar funding gap reached around \$1.2 trillion prior to the GFC. In my model, this is equivalent to setting dollar shortages $(1 + \gamma)L^* - A^*/R^*$ to be 15% of total dollar liabilities, L^* .

The parameters of the model are calibrated to match this data. I follow [Gabaix and Maggiori \(2015\)](#) in setting $\omega = \omega^* = 0.1$ so that non-tradables account for 90% of the consumption basket. I set $\beta = \beta^* = 0.985$ which are relevant to match the interest rates of $R = 1.015$ and $R^* = 1.013$, quarterly. The financial friction is set to $\gamma = 0.64$. The rest of the parameters are set to make the two countries broadly similar: $\eta_1 = 0.47$, $\eta_2 = 0.5$, $Y_1^N = 2.58$, $Y_1^{*N} = 2.55$, $Y_2^N = Y_2^{*N} = 2.5$, $A = .07$, $L = .04$, $A^* = .05$, $L^* = .03$. I keep

Table 1: Targeted variables

Variable	Description	Target	Model
$\frac{e^H - e^L}{e^L}$	ER depreciation	12.5%	12.5%
	\$ shortage (%)	15%	15%
R^*	US interest rate	1.013	1.013
R	EU interest rate	1.015	1.015

Table 2: Untargeted variables

Variable	Description	Data	Model
$\frac{A^*}{A^* + Y_2^{*N}}$	US output loss	2.2%	2.0%
$\frac{A}{A + Y_2^N}$	EU output loss	1.8%	2.9%

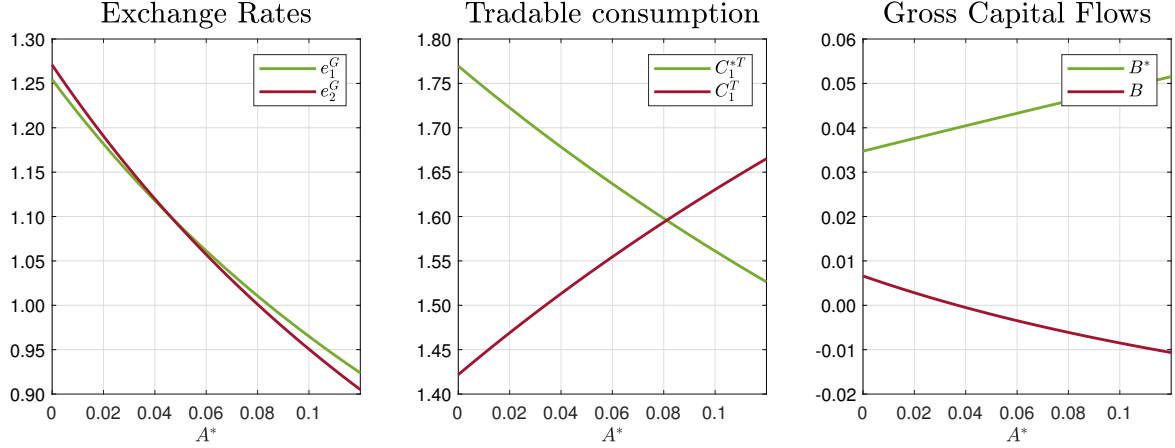


Figure 6: Key variables in the “good” equilibrium

$\Psi = \Psi^* = 0$ and $\Omega = \Omega^* = 0$ for now.

The results of this exercise are shown in Table 1. This simple model is able to match the behavior of key variables around the GFC, such as the output decline in the EU, the dollar appreciation with respect to the euro, and ex-ante interest rates in both economies. Some relevant untargeted variables such as the drop in EU and US output²⁵ respond in the expected direction, but they react slightly more drastically in the model compared to what the data suggests²⁶.

Finally, based on this simple calibration, Figure 6 shows how exchange rates, consumption, and gross capital flows respond to changes in A^* in the “good” equilibrium. In line with the intuition, higher returns on US assets raise bank profits and thus EU households’ relative wealth, leading to a euro appreciation (i.e., a lower exchange rate) in both periods. Tradable consumption shifts accordingly: C_t^T rises with stronger EU demand, while C_t^{*T} falls. Gross capital flows adjust consistently with these dynamics.

²⁵The quarterly output drop in the US during Q4-2008 was 2.2%, while it was 1.8% for the EU.

²⁶Online Appendix A5 presents an extension of the model where households’ have CES utility functions. By having more flexibility to calibrate the different elasticities in the households’ problem, the exchange rate reacts more drastically to changes in consumption, which brings the model’s results closer to the data.

5 Lending of last resort

In this section I introduce a government in each economy that intervenes in financial markets in period 1, discuss the motives behind these interventions, and find under what conditions governments can prevent the collapse of global banks.

An economy that is exposed to a “bad” equilibrium driven by pessimistic expectations could usually benefit from the intervention of a benevolent government, a social planner, or in this case, a lender of last resort. I follow [Bocola and Lorenzoni \(2020\)](#) and [Gertler and Kiyotaki \(2015\)](#) in modelling the lender of last resort and introduce a government that can make a transfer S to global banks in period $t = 1$. This transfer is financed by imposing a tax τ on consumers’ endowment of non-tradables Y_1^N , which is later transferred back to the households with interests²⁷ R^S (non-distortionary tax).

Intuitively, the intervention is successful if the lender of last resort has the capacity to provide the liquidity that banks need, so that households rule out the possibility of a banking collapse from their expectations, and are willing to provide banks with deposits. As it is common in these type of models, the intervention might not need to materialize, as long as the government can convince the markets that its commitment to prevent the collapse scenario is credible ([Céspedes et al., 2017](#)). Naturally, the credibility of this claim depends on the resources that the government can access.

5.1 Intervention by the ECB

Consider first the case where the central bank in the EU (ECB) acts as the lender of last resort to global banks. This is a starting scenario, where a central bank tries to bail out domestic banks and avoid a collapse of the domestic financial system. For now, I will not motivate this intervention with potential welfare gains, but assume that it is part of the central bank mandate to avoid a financial crisis.

Recall that for simplicity, I set $L = 0$ so that all the initial debt held by banks is denominated in dollars (L^*). The ECB then sets R^S and transfers S to banks such that their profits are

$$\Pi = e_2 A^* + A - R^S S ,$$

meaning that the full amount of the initial liabilities in dollars is covered with the transfer in euros,

$$e_1 L^* = S , \quad \text{where } S = \tau Y_1^N \tag{21}$$

²⁷I will not focus on how the interest rate is set, but simply assume that the central bank charges the same interest rate as households would, had they decided to provide the funding.

Finally, equation (21) shows that the size of the intervention $\tau Y_1^N = e_1 L^*$ depends, crucially, on the exchange rate. A stronger dollar means that the amount of euros that the ECB needs to cover the initial dollar liabilities from global banks is larger. Naturally, the intervention is limited by the amount of resources in the economy, which in this case is given by Y_1^N . On top of that, I follow [Bocola and Lorenzoni \(2020\)](#) and assume that fiscal capacity is limited²⁸ in the following way.

Assumption 2. *There is an upper bound on the tax rate that the government can apply for this intervention, such that $\tau \leq \bar{\tau}$.*

As I mentioned previously, in order for the intervention to be successful, agents must believe that the lender of last resort has enough resources to prevent the “bad” equilibrium at all costs. In this framework, that means that the ECB must have enough tax income²⁹ to cover the banks’ dollar liabilities, even in the state of the world where the dollar is largely appreciated (in other words, when the exchange rate is e_1^B). This comes from the fact that, when a central bank intervenes, it takes the exchange rate as given, even though -eventually- its actions will affect this variable. Considering equation (21) and the tax limit, the next proposition captures this insight.

Proposition 3. *Consider the ECB sets transfers S in euros to cover banks’ dollar liabilities L^* and that Assumption 2 holds. These transfers are financed with taxes on EU households’ non-tradable endowment such that $S = \tau Y^N$. The intervention will eliminate the “bad” equilibrium if it is credible, which happens when the following condition holds:*

$$\bar{\tau} Y^N > e_1^B L^* = \frac{\eta^*}{\eta} L^* .$$

Moreover, if the commitment to intervene is credible, the ECB would not have to intervene to prevent the collapse scenario.

Proof. In Appendix [B.6](#) □

A graphic illustration of the previous proposition is presented in Figure 7. If a fixed tax limit is considered, it is possible to analyze how the fundamentals of the global economy might give rise to unpreventable equilibria, from the perspective of the ECB. Denote $e^{\bar{\tau}}$ as the maximum exchange rate that the central bank can handle, given $\bar{\tau}$. Consider a “bad” equilibrium such as the one given by the blue and solid red line. Since the exchange rate during a collapse (e^B) is lower than $e^{\bar{\tau}}$, the ECB can effectively prevent the financial crisis from materializing, as shown by the dotted green. Now, for instance, if

²⁸This can be motivated in many ways. From the point of view of a central bank, this limit could represent a maximum level of inflation that can be tolerated given the massive liquidity injection, or an upper bound to the potential losses that the bank can take given a (very) low default risk.

²⁹Online Appendix [A4.2](#) shows that the intervention by the ECB might still be unfeasible even if it could tax and transfer tradable goods to global banks.

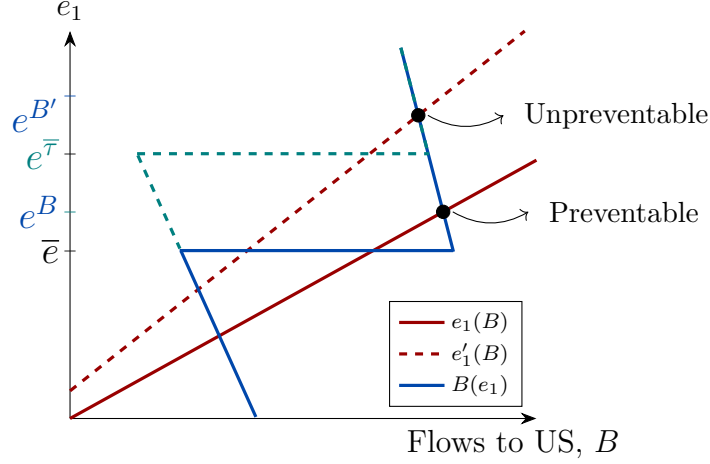


Figure 7: Equilibria under ECB intervention

the endowment of tradables goods (η_1) in the EU is lower, relative prices in that economy will be higher, which leads to an increase in the exchange rate in both equilibria, as shown by the red dotted line. The limitations of the central bank make a scenario with $e^{B'} > e^{\bar{\tau}}$ unpreventable.

5.2 Intervention by the Fed

Consider now the intervention from the Fed instead of the ECB. In the model, the motivation for the Fed to intervene will come mainly from preventing a collapse of productive investments in the US and a subsequent decline in US non-tradable output in period 2, but a more comprehensive analysis of the welfare implications is left for the next section. The mechanism to intervene is the same as the one described before, but now the Fed is the one transferring resources S^* directly to global banks³⁰. This transfer is financed with taxes τ^* on US households' non-tradable output (recall that $Y_1^N = Y_1^{*N}$). An important difference between these two central banks is that one provides euros (EU non-tradable goods), while the other provides dollars (US non-tradable goods). The Fed then transfers S^* dollars to cover banks' dollar liabilities, such that

$$L^* = S^* , \quad \text{where } S^* = \tau^* Y^{*N} . \quad (22)$$

Equation (22) shows that, unlike the case for the ECB, the size of the intervention $Y^{*N} \tau^* = L^*$ by the Fed does not depend on the exchange rate. This is a key difference with any other central bank in the world. When banks operate, we have that $e_1^G < 1$ so one unit of EU non-tradable goods has more value than one unit of US non-tradable goods, i.e. one *euro* is worth more than one *dollar*.

³⁰In practice the transfer from the Fed goes to the foreign central bank, which eventually distributes the resources to the domestic banks. However, in the absence of additional frictions, this would be equivalent to the Fed directly helping foreign banks.

Nevertheless, during a financial crisis, the situation changes. Whenever banks go bust and the exchange rate appreciates to $e_1^B > 1$, the dollar is stronger than the euro. Again, this is consistent with the evidence shown in Section 2 suggesting a large appreciation of the dollar during a crisis, and is also in line with the “dash-for-dollars” (Cesa-Bianchi and Eguren-Martin, 2021), “flight-to-safety” (Kekre and Lenel, 2024), or “scrambling-for-dollars” (Bianchi et al., 2021) phenomena, in which the demand for dollars increase during turbulent episodes. The implications of a weaker euro for the ECB are that now the required intervention is larger than under “good” times. Meanwhile, the required size of the Fed’s intervention remains unchanged. To compare the Fed’s and the ECB’s interventions, I will further assume the following.

Assumption 3. *Both governments face the same tax limit, such that $\tau = \tau^* \leq \bar{\tau}$.*

Considering this, a very particular case might arise: one in which the Fed has the resources to engineer a credible intervention, while the ECB does not. The next proposition summarizes these results and the conditions for this to happen.

Proposition 4. *Consider that Assumption 3 holds, countries receive the same amount of non-tradable endowments $Y_1^N = Y_1^{*N}$, and that the exchange rate during a financial crisis is $e_1^B > 1$. To be effective, the intervention from the ECB requires setting $\tau Y^N \equiv e_1^B L^*$, which is higher than the required tax rate that the Fed has to impose $\tau^* Y^N = L^*$. Moreover, only the Fed will be able to eliminate the “bad” equilibrium, if the following condition holds:*

$$\underbrace{\frac{\eta^*}{\eta} L^*}_{\text{Liq. needs in euros}} > \underbrace{\bar{\tau} Y^N}_{\text{Maximum intervention}} > \underbrace{L^*}_{\text{Liq. needs in dollars}}$$

Proof. In Appendix B.7 □

A graphic illustration of this proposition is presented in Figure 8. For any level of exchange rate in the “bad” equilibrium that is below the limit $\bar{\tau} Y^N / L^*$, both the Fed and the ECB can intervene credibly. However, if e_1^B is higher than that limit, we enter a zone in which only the Fed has the resources to prevent a financial crisis.

These results provide a theoretical explanation—in a very reduced form—as to why the Fed provided the required liquidity to non-US banks during the GFC and the Covid-19 crisis, and not the corresponding domestic central banks. In practice, such a massive intervention would have imposed significant costs on them and strained fiscal resources during periods of economic turbulence. Additionally, it is also reasonable to think that an injection of euros from the ECB to bail out the struggling banks could have triggered an even larger depreciation with respect to the dollar, amplifying the initial shock.

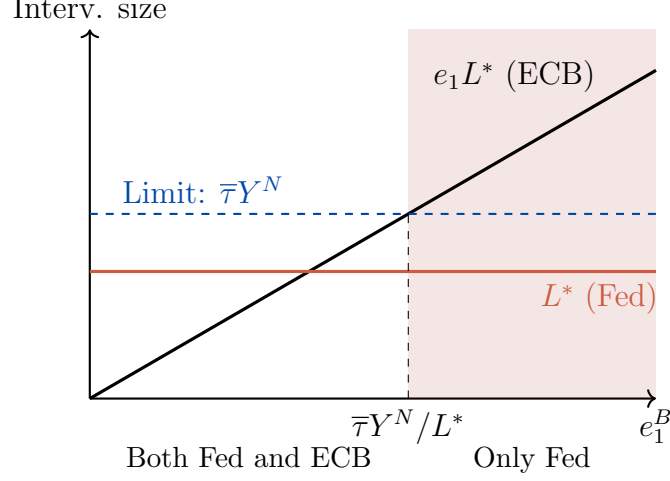


Figure 8: Intervention by the Fed and the ECB

6 Incentives and welfare implications

So far, this paper has described the mechanism through which governments or central banks can bail out global banks, without much discussion about the incentives behind these interventions. I now turn to this crucial aspect by analyzing the welfare implications of converging to each of the stable equilibria featured in the model.

This section introduces a more general version of the model, relaxing several assumptions imposed in Section 3 for ease of exposition. I do so for two main reasons: (i) to demonstrate that the core results are robust to less restrictive assumptions regarding banks' funding structures, fire-sale dynamics, and repayment capacities; and (ii) to better highlight the general equilibrium forces driving the Fed's decision to supply dollar liquidity to the world.

In particular, I now allow for euro-denominated debt to be rolled over and for the liquidation process to be only partially unproductive. Formally, I let

$$L > 0, \quad \Psi, \Psi^* > 0, \quad \Omega^* > 0,$$

which from equation (13), implies that $\Omega > 0$ for certain values of the exchange rate³¹. All other simplifying assumptions presented in Section 3.4 are maintained, except the normalization $Y_1^N = Y_1^{*N} = 1$. Notice that in this generalized setting, when global banks collapse, they manage to recover a fraction of their long-term assets through fire sales and use the proceeds to partially repay creditors. This setup more realistically reflects crisis episodes, where banks sell assets at a discount when unable to roll over their liabilities.

The exchange rate in the “good” equilibrium remains unchanged under these general-

³¹This occurs when $\Psi A > e_1(\Omega^* L^* - \Psi^* A^*)$.

izations³². What changes is the characterization of the “bad” equilibrium, as partial asset recovery and repayment affect both banks’ balance sheets and the cross-country wealth distribution. The full derivation is presented in Online Appendix B.2. The equilibrium exchange rate in the collapse scenario is now given by

$$e_1^B = \frac{\eta^*}{\eta - \kappa \cdot \underbrace{\frac{\Omega^* L^* - \Psi^* A^*}{Y_1^{*N} + \Psi^* A^*}}_{\text{Wealth transfer}}} \cdot \frac{Y_1^N + \Psi A}{Y_1^{*N} + \Psi^* A^*} , \quad (23)$$

where $\kappa \equiv \frac{1-\omega}{\omega} \frac{1}{1+\beta}$. Equation (23) shows that the exchange rate depends primarily on relative consumption of non-tradable goods, $(Y_1^N + \Psi A)/(Y_1^{*N} + \Psi^* A^*)$, and the distribution of tradable endowments across countries, η^*/η . It also highlights the role of cross-border wealth transfers. In particular, the term $\Omega^* L^* - \Psi^* A^*$ captures the gap between the repayment of dollar liabilities and the liquidation value of dollar assets. I assume this gap is positive, i.e. $\Omega^* L^* - \Psi^* A^* > 0$, so that some of the proceeds from liquidating euro-denominated assets must be used to cover dollar obligations. As a result, e_1^B is relatively higher because a larger wealth transfer from the EU to the US boosts aggregate demand in the US, and puts upward pressure on the dollar.

A new endogenous variable in the model is the repayment ratio on euro liabilities, Ω . From equation (13), it can be expressed as a function of the exchange rate,

$$\Omega = \frac{1}{L} \left(\Psi A - e_1 (\Omega^* L^* - \Psi^* A^*) \right) . \quad (24)$$

Differentiating with respect to e_1 yields

$$\frac{\partial \Omega}{\partial e_1} = - \frac{\Omega^* L^* - \Psi^* A^*}{L} < 0 ,$$

showing that, since the liquidation value of dollar assets is not sufficient to cover the partial repayment of dollar liabilities, a higher exchange rate implies that more euro-denominated assets must be used to cover the dollar shortfall, leaving less available to repay euro liabilities.

6.1 Consequences of a financial crisis

Denote with a subscript G variables in the “good” equilibrium, and with B those in the “bad” one. Welfare losses from the liquidation of global banks are given by the difference

³²Since banks are owned by EU households and transfer any potential profits to them, the size of the euro-denominated deposits L becomes irrelevant when determining the equilibrium exchange rate in this scenario.

between the utility of households in both scenarios:

$$U_G^* - U_B^* = \sum_{t=1}^2 \beta^{*t-1} \left\{ (1 - \omega) \underbrace{\left(\ln C_{G,t}^{*N} - \ln C_{B,t}^{*N} \right)}_{NT^* \text{ goods}} - \omega \underbrace{\left(\ln C_{B,t}^{*T} - \ln C_{G,t}^{*T} \right)}_{T^* \text{ goods}} \right\}, \quad (25)$$

with $U^* = \ln(C_1^*) + \beta \ln(C_2^*)$ and equivalently for the EU (without $*$).

The welfare costs of a financial crisis can be decomposed into two components. The first arises from the forced liquidation of productive investments. If held to maturity, these investments would have increased the supply of non-tradable goods in period 2. Instead, only a fraction of their value is recovered early through fire sales. As a result, C_2^N and C_2^{*N} fall by A and A^* , while C_1^N and C_1^{*N} increase only by ΨA and $\Psi^* A^*$, respectively. These direct effects reduce welfare in both countries³³ and are captured by the first term in (25).

The second component concerns tradable consumption and stems from financial losses and wealth redistribution. Households in both countries lose part of the deposits they initially held with these banks, while EU households also give up the potential profits that global banks would have earned. From a partial equilibrium perspective, these effects are unambiguously negative for both economies. However, general equilibrium forces—particularly those shaping the distribution of wealth and tradable consumption—can tilt the balance in a different direction. As Section 4 showed, when a crisis hits, capital flows to the US increase, and the dollar appreciates. This allows US households to consume more tradables. Conversely, the relative wealth of EU households drops from the collapse of global banks, which limits the amount of tradable goods they can consume. These effects are captured by the second term in (25). Importantly, they might reduce welfare losses for US households, but amplify them in the case of EU households.

To fully understand the strength of these general equilibrium effects and how they impact consumption, it is worth decomposing C_1^{*T} as follows:

$$C_1^{*T} = Y_1^{*T} - \underbrace{\frac{1}{p_1^*}}_{\text{Price effect}} \cdot \underbrace{\mathcal{B}^*}_{\text{Flows effect}} + \underbrace{\frac{1}{p_1^*} \cdot \mathcal{L}^*}_{\text{Deposits effect}} \quad (26)$$

Equation (26) shows that in case of a collapse, $\mathcal{L}^* = \Psi^* L^*$ and therefore the loss of a fraction $1 - \Psi^*$ of the deposits reduces the disposable income that US households can allocate to consumption. But on the other hand, they experience larger capital inflows ($\downarrow \mathcal{B}^* < 0$), and lower relative prices ($\downarrow p_1^*$) from the appreciation of the dollar. Overall, these two effects lead to an increase in C_1^{*T} . In period 2, C_2^{*T} increases as well during

³³The term $\sum_{t=1}^2 \beta^{*t-1} (1 - \omega) \left(\ln C_{G,t}^{*N} - \ln C_{B,t}^{*N} \right)$ boils down to $(1 - \omega) \left(\ln \frac{Y_1^{*N}}{Y_1^{*N} + \Psi^* A^*} + \beta \ln \frac{Y_2^{*N} + A^*}{Y_2^{*N}} \right)$. I consider Ψ and Ψ^* small enough so that the term is positive (welfare losses) for both countries.

a crisis, mostly because of the drop in interest rates R^* .

If we put these effects together, it is possible to draw some interesting conclusions. On one hand, preventing the collapse of EU-owned global banks is always beneficial for the EU, since they consume fewer non-tradable and tradable goods in the “bad” equilibrium, compared to the “good” one. On the other hand, the US faces two opposite forces going in different directions. US households are negatively affected by the loss of non-tradable goods, but this is mitigated by the gain from higher consumption of tradable goods, coming from lower relative prices and a stronger dollar due to weaker demand in the EU.

6.2 Trade-off for the Fed

Whether US households experience an overall welfare gain or loss when global banks collapse depends on the balance of the two opposing forces in (25). The relative strength of these forces is determined by the fundamentals of the world economy, the size and composition of global banks’ balance sheets, and the extent of asset recovery in a crisis. The following proposition collects the conditions that determine the welfare implications for the US.

Proposition 5. *Assume $\beta = \beta^* = 1$. Comparing the utility obtained by households under the “good” and the “bad” equilibria, EU households always experience a welfare loss ($U_G - U_B > 0$). On the contrary, US households might benefit from higher tradable consumption, but face lower consumption from non-tradable goods. Overall, the Fed will lack the incentives to intervene and provide the liquidity required to foreign global banks if $U_G^* - U_B^* < 0$, which happens if:*

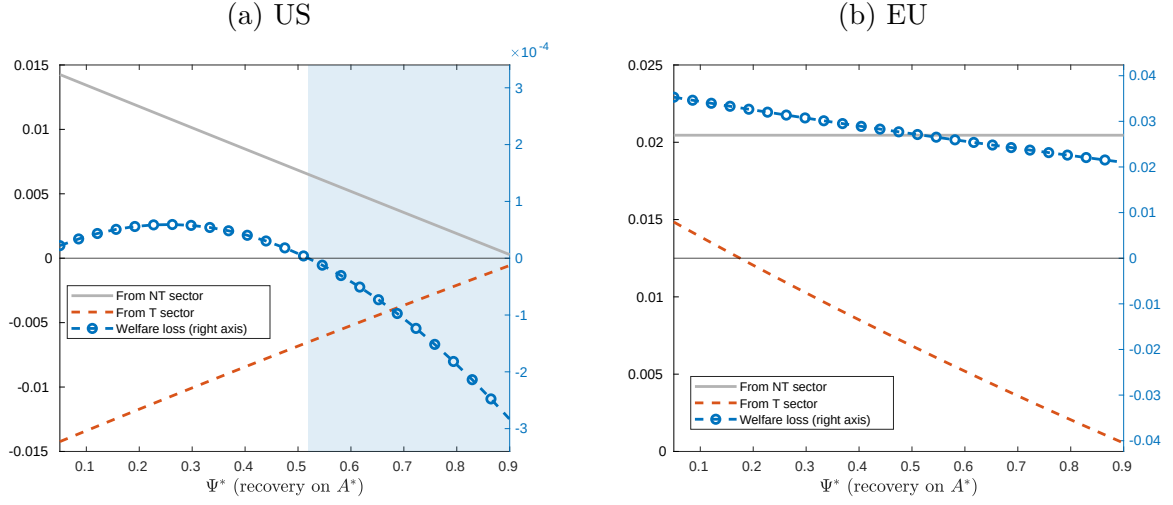
$$\left(\frac{1 + \frac{A^*}{Y_2^{*N}}}{1 + \frac{\Psi^* A^*}{Y_1^{*N}}} \right)^\phi < \frac{1 + \phi \left(\frac{A^*}{Y_2^{*N} + A^*} - \frac{L^*}{Y_1^{*N}} \right)}{1 - \phi \left(\frac{\Omega^* L^* - \Psi^* A^*}{Y_1^{*N} + \Psi^* A^*} \right)}$$

where $\phi \equiv \frac{1-\omega}{2\omega} > 0$.

Proof. In Appendix B.8 □

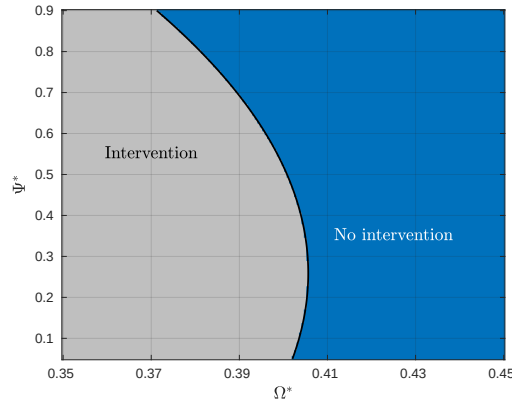
The inequality contrasts the relative increase in non-tradable output across equilibria on the left-hand side with the magnitude of cross-border wealth transfers on the right-hand side. Two parameters are key. First, the repayment rate on US deposits in the collapse state (Ω^*) monotonically reduces the Fed’s incentive to bail out foreign banks, as a higher Ω^* means US households recover more in the “bad” equilibrium, enlarging the wealth transfer towards the US. Second, the fire-sale recovery rate on dollar assets, Ψ^* , works in opposite directions. A higher Ψ^* preserves more productive assets and thus more US non-tradable supply in a crisis (weakening the case for intervention), but it simultaneously shrinks the wealth transfer from the EU to the US because fewer euro-asset proceeds

Figure 9: Welfare losses



Note: Considers the parameter values described in Section 4, with $\Psi = 0.2$ and $\Omega^* = 0.4$. The light-blue area in panel (a) represents the values of Ψ^* for which the Fed decides not to intervene.

Figure 10: Pairs of Ψ^* and Ω^* and Fed's incentives to intervene



Note: The blue area represents the combination of parameters for which the Fed decides not to intervene.

are needed to cover dollar deposits $\Omega^* L^*$ (strengthening the case for intervention). Hence, the net effect of Ψ^* on the Fed's incentives is, in general, ambiguous.

To clarify the trade-off, I provide a simple numerical example where I compute welfare using (25) for both countries, under the calibration from Section 4, allowing repayment and recovery rates to vary. To isolate the role of Ψ^* , I first hold Ω^* and Ψ constant. Figure 9 shows that EU households always experience welfare losses, but they are decreasing in Ψ^* . The reason is that higher recoveries of dollar assets mean that fewer capital flows need to go from the EU to the US to cover for the repayment of US liabilities, allowing EU households to fund more consumption of tradable goods.

For the US, the trade-off is evident: losses in the non-tradable component rise with Ψ^* , while gains in tradable consumption also increase. The overall effect on US welfare is

thus ambiguous. Under this calibration, for a sufficiently low Ψ^* , the capital-inflow gains can dominate, weakening the incentive to provide dollar liquidity.

Finally, I examine the sensitivity of US welfare losses to Ω^* and its interaction with Ψ^* . Figure 10 shades in blue the region of the parameter space (Ψ^*, Ω^*) for which a crisis raises US welfare— that is, where moving from the “bad” to the “good” equilibrium would lower US households’ utility. Consistent with the intuition above, the Fed’s incentives to intervene are weakest when dollar assets can be liquidated with relatively small discounts (high Ψ^*) and when US deposit recoveries are large (high Ω^*), although some non-linearities are also present.

7 Conclusions

In this paper I develop a framework to study the global macroeconomic implications of the Fed’s swap lines to foreign central banks in times of crisis. Non-US global banks act as “bankers of the world” by intermediating flows between the US and the rest of the world in their respective currencies, and investing in dollar assets. However, given maturity mismatches in dollars and financial constraints, they can be exposed to exchange rate fluctuations. Therefore, a significant dollar appreciation could lead to a banking crisis, generating a drop in the aggregate demand and a further currency depreciation in the rest of the world. I argue that this mechanism opens the door to self-fulfilling crises driven by pessimistic expectations.

In this context, the world economy can benefit from a lender of last resort. However, in a state of the world where the dollar is strong relative to other currencies, and given the size of global banks’ balance sheets, non-US central banks without significant dollar reserves might lack the resources to prevent the “bad” equilibrium. The Fed, on the other hand, can intervene by providing dollar liquidity directly. Nevertheless, its incentives to bail out foreign global banks might not be in line with the interests of the rest of the world. The reason is that there are general equilibrium forces at play that could benefit the US and mitigate the consequences of a global financial crisis on their economy.

I believe this framework represents a useful starting point to think about the macroeconomic implications and incentives around the US as the international lender of last resort. However, there are still many aspects left to explore. An exciting research avenue is to understand the moral hazard issues that could arise from such an intervention³⁴, not only for the US, but also for the receiving countries. Effectively addressing these issues is crucial for assessing the future of the dollar’s global dominance, the risks that threaten it, and identifying the steps the US can take to mitigate those risks to maintain confidence in the dollar.

³⁴Online Appendix A6 sheds light on this issue, and how it matters for the main results of the paper.

Appendix

A Exchange rate and dollar shortages

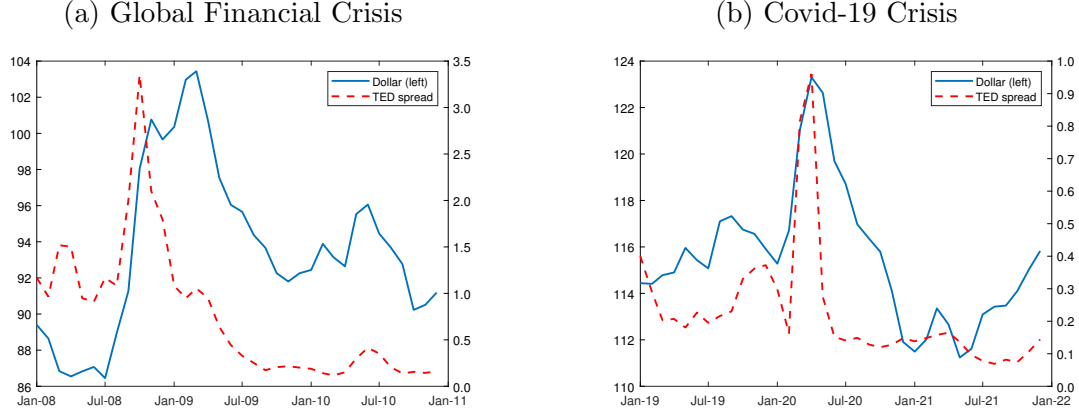


Figure 11: Dollar (index) and TED spread (%)

Note: The TED spread is defined as the difference between the 3-month LIBOR rate and the 3-month T-bill rate. An increase can be understood as an increase in the interest rates that banks have to pay to access dollar funding in the international markets. **Source:** Fed.

B Proofs and derivations

B.1 Derivation of Equation 19

From the households' optimality conditions we obtain that $p_t C_t^T = \frac{\omega}{1-\omega} C_t^N$ and $p_t^* C_t^{*T} = \frac{\omega}{1-\omega} C_t^{*N}$. Now consider the tradable market clearing condition,

$$C_t^T + C_t^{*T} = Y_t^T + Y_t^{*T}$$

and multiply both sides of the equation by p_t . Combining all these expression, the market clearing condition for non-tradable goods, and the law of one price $e_t p_t^* = p_t$, we get the following expressions for the price of tradable goods:

$$p_t = \frac{\omega}{1-\omega} \frac{1}{Y_t^T + Y_t^{*T}} \left[e_t C_t^{*N} + C_t^N \right]. \quad (27)$$

Finally, using the market clearing conditions in $t = 1$, $C_1^N = Y_1^N$ and $C_1^{*N} = Y_1^{*N}$, the simplifying assumption $Y_1^{*N} = Y_1^N = 1$, and combining (27) with the households' optimality condition and the trade balance in period 1 given by $p_1(Y_1^T - C_1^T) = B$, we

get

$$e_1 = \frac{\eta_1^*}{\eta_1} + B \frac{1-\omega}{\omega} \frac{1}{\eta_1},$$

where $\eta_1 \equiv \frac{Y_1^T}{Y_1^T + Y_1^{*T}}$ and $\eta_1^* \equiv 1 - \eta_1$.

B.2 Derivation of Equation 23

When banks shut down, $\mathcal{L} = \Omega L$ and $\mathcal{A} = \Psi A$. From market clearing in the non-tradable sector, $Y_1^N - C_1^N = \Psi A$. Using the EU households' budget constraint, we obtain

$$\Omega L + p_1(Y_1^T - C_1^T) - \Psi A = \frac{e_1}{e_2 R^*} p_2(C_2^T - Y_2^T). \quad (28)$$

Substituting (27) and the intratemporal condition $p_t C_t^T = \frac{\omega}{1-\omega} C_t^N$, equation (28) can be rewritten as

$$\Omega L + \frac{\omega}{1-\omega} \eta_1 (e_1 C_1^{*N} + C_1^N) - \frac{\omega}{1-\omega} C_1^N - \Psi A = \frac{e_1}{e_2 R^*} \left(\frac{\omega}{1-\omega} C_2^N - \frac{\omega}{1-\omega} \eta_2 (e_2 C_2^{*N} + C_2^N) \right).$$

Next, applying the intertemporal Euler condition for EU households (when banks do not operate),

$$\frac{p_1 C_1^T}{p_2 C_2^T} \beta = \frac{e_1}{e_2 R^*},$$

together with market clearing in non-tradables and assuming $\eta_t = \eta$ for all t , yields the following expression for e_1 and Ω :

$$e_1 = \frac{\eta^* - \frac{1-\omega}{\omega} \frac{\Omega L - \Psi A}{Y_1^N + \Psi A}}{\eta} \cdot \frac{Y_1^N + \Psi A}{Y_1^{*N} + \Psi^* A^*}.$$

Finally, using the repayment condition (13),

$$\Omega L - \Psi A = e_1 (\Psi^* A^* - \Omega^* L^*),$$

and substituting back gives the exchange rate in the collapse equilibrium:

$$e_1^B = \frac{\eta^*}{\eta - \kappa \frac{\Omega^* L^* - \Psi^* A^*}{Y_1^{*N} + \Psi^* A^*}} \cdot \frac{Y_1^N + \Psi A}{Y_1^{*N} + \Psi^* A^*},$$

where $\kappa \equiv \frac{1-\omega}{\omega} \cdot \frac{1}{1+\beta}$.

B.3 Proof of Lemma 3.1

Lemma 3.1. *Suppose that Assumption 1 holds. A necessary condition for all banks to operate in equilibrium is,*

$$e_1 \leq \frac{A/R - (1 + \gamma)L}{(1 + \gamma)L^* - A^*/R^*} \equiv \bar{e},$$

where $R = (Y_2^N + A)/\beta Y_1^N$ and $R^* = (Y_2^{*N} + A^*)/\beta^* Y_1^{*N}$.

Proof. Let us consider condition (10) expressed in terms of e_1 ,

$$\frac{A}{R} + e_1 \frac{A^*}{R^*} \geq (1 + \gamma)(L + e_1 L^*)$$

It is straightforward to see that an increase in e_1 will increase both the left-hand-side (LHS) and the right-hand-side (RHS) of the previous inequality. However, under Assumption 1, the LHS increases at a slower rate (A^*/R^*) than the RHS ($(1 + \gamma)L^*$). Therefore, $\exists e_1$ large enough such that the inequality no longer holds.

Next, if we further assume that $A/R > (1 + \gamma)L$, that value is positive. Combining these two facts, we can conclude that all banks will be able to operate only if e_1 is below a certain threshold. $\square$

B.4 Proof of Proposition 1

Proposition 1. *Suppose that Assumption 1 holds, $\eta_t = \eta \forall t$, and let $\bar{e}$ be the value of e_1 that makes condition (10) hold with equality. Then, multiple equilibria are possible if*

$$\underbrace{\frac{\eta^*}{\eta + \frac{1}{1+\beta} \frac{1-\omega}{\omega} (A^*/R^* - L^*)}}_{e^G} \leq \underbrace{\frac{A/R}{(1 + \gamma)L^* - A^*/R^*}}_{\bar{e}} \leq \underbrace{\frac{\eta^*}{\eta}}_{e^B}$$

where $R = (A + Y_2^N)/\beta Y_1^N$ and $R^* = (A^* + Y_2^{*N})/\beta Y_1^{*N}$.

Proof. Let us consider the first inequality. It follows from the proof of Lemma 3.1 that $e^G < \bar{e}$ is a necessary condition for the “good” equilibrium to exist. The second inequality states that $\bar{e} < e^B$ for the “bad” equilibrium to exist.

Assume that such equilibrium exists even if $e^B < \bar{e}$. In that case, and given that households have perfect foresight, it must be that they expected e^B , and decided not to provide the funds to global banks, leading to their collapse. However, this contradicts households’ rationality. The reason is that, since they use the exchange rate as a coordination device, if they expected an exchange rate that would not violated the incentive compatibility constraint of banks, they would have given them the funds they

need, avoiding the collapse. It follows that $\bar{e} < e^B$ in order for the “bad” equilibrium to exist. $\square$

B.5 Proof of Proposition 2

Proposition 2. *If there are three equilibria and we compare the two stable ones, we obtain the following predictions about the crisis equilibrium with respect to the standard equilibrium:*

- i. *The dollar is more appreciated;*
- ii. *Banks face tighter financial conditions and struggle to roll over their debt;*
- iii. *Global output and wealth in the EU are lower;*
- iv. *Net capital flows to the US are larger.*

Proof. The proof for each item in the proposition will be provided separately.

- i. Follows from the conditions in Proposition 1: $e_1^B > e_1^G$.
- ii. Follows from the fact that banks collapse in the “bad” equilibrium.
- iii. When banks do not operate, non-tradable output in the US is simply given by the endowments in both periods, $Y_1^{N*} + Y_2^{N*}$. On the contrary, if US assets owned by global banks materialize, non-tradable output in the US increases to $Y_1^{N*} + Y_2^{N*} + A^*$. The equivalent occurs in the EU. As for wealth in the EU, they experience higher relative prices ($e_1^B > e_1^G$) and they lose banks profits $\Pi > 0$ when a crisis hits. This represents lower relative wealth.
- iv. Consider equation (19) and rearrange it in terms of e_1 ,

$$B = \frac{\omega}{1 - \omega} (\eta_1(1 + e_1) - 1) .$$

From the previous equation, since $e_1^B > e_1^G$, it must be that $B^B > B^G$, meaning that capital flows to the US in the “bad” equilibrium are larger than in the “good” one. $\square$

B.6 Proof of Proposition 3

Proposition 3. *Consider the ECB sets transfers S in euros to cover banks’ dollar liabilities L^* and that Assumption 2 holds. These transfers are financed with taxes on EU households’ non-tradable endowment such that $S = \tau Y^N$. The intervention will eliminate the “bad” equilibrium if it is credible, which happens when the following condition holds:*

$$\bar{\tau}Y^N > e_1^B L^* = \frac{\eta^*}{\eta} L^* .$$

Moreover, if the commitment to intervene is credible, the ECB would not have to intervene to prevent the collapse scenario.

Proof. The liquidity needs from global banks $e_1 L^*$ have to be cover by euro transfers from the ECB, thus

$$e_1 L^* = S \quad (29)$$

Moreover, these transfers are funded by taxes on EU households non-tradable endowment, thus

$$\tau Y_1^N = S \quad (30)$$

Combining (29) and (30), we get that $e_1 L^* = \tau Y_1^N$. Since τ is increasing in e_1 , and given the upper bound on the tax rate, $\tau < \bar{\tau}$, the intervention will eliminate the “bad” equilibrium if it is credible, which happens when the following condition holds, $\bar{\tau} > e_1^B L^* / Y_1^N$. $\square$

B.7 Proof of Proposition 4

Proposition 4. *Consider that Assumption 3 holds, countries receive the same amount of non-tradable endowments $Y_1^N = Y_1^{*N}$, and that the exchange rate during a financial crisis is $e_1^B > 1$. To be effective, the intervention from the ECB requires setting $\tau Y^N \equiv e_1^B L^*$, which is higher than the required tax rate that the Fed has to impose $\tau^* Y^N = L^*$. Moreover, only the Fed will be able to eliminate the “bad” equilibrium, if the following condition holds:*

$$\underbrace{\frac{\eta^*}{\eta} L^*}_{\text{Liq. needs in euros}} > \underbrace{\bar{\tau} Y^N}_{\text{Maximum intervention}} > \underbrace{L^*}_{\text{Liq. needs in dollars}}$$

Proof. To be effective, the intervention from the ECB requires setting $\tau = e_1^B \frac{L^*}{Y_1^N}$, while the Fed requires setting $\tau^* = \frac{L^*}{Y_1^N}$. Since $e_1^B = \eta^* / \eta > 1$, then $\tau = e_1^B \tau^* > \tau^*$. $\square$

B.8 Proof of Proposition 5

Proposition 5. *Assume $\beta = \beta^* = 1$. Comparing the utility obtained by households under the “good” and the “bad” equilibria, EU households always experience a welfare loss ($U_G - U_B > 0$). On the contrary, US households might benefit from higher tradable consumption, but face lower consumption from non-tradable goods. Overall, the Fed will*

lack the incentives to intervene and provide the liquidity required to foreign global banks if $U_G^* - U_B^* < 0$, which happens if:

$$\left(\frac{1 + \frac{A^*}{Y_2^{*N}}}{1 + \frac{\Psi^* A^*}{Y_1^{*N}}} \right)^\phi < \frac{1 + \phi \left(\frac{A^*}{Y_2^{*N} + A^*} - \frac{L^*}{Y_1^{*N}} \right)}{1 - \phi \left(\frac{\Omega^* L^* - \Psi^* A^*}{Y_1^{*N} + \Psi^* A^*} \right)}$$

where $\phi \equiv \frac{1-\omega}{2\omega} > 0$.

Proof. EU households' welfare is given by the consumption of tradable and non-tradable goods in both periods:

$$\mathcal{U} = (1 - \omega) \ln(C_1^N) + \omega \ln(C_1^T) + \beta(1 - \omega) \ln(C_2^N) + \beta\omega \ln(C_2^T)$$

Using the market clearing conditions, and the households' first order condition $C_t^T = C_t^N \frac{\omega}{1-\omega} \frac{1}{p_t}$, the welfare loss is given by

$$\begin{aligned} U_G - U_B &= (1 - \omega) \left\{ \underbrace{\ln \left(\frac{Y_1^N}{\Psi A + Y_1^N} \right) + \beta \ln \left(\frac{A + Y_2^N}{Y_2^N} \right)}_{NT \text{ goods}} \right\} - \underbrace{\omega \sum_{t=1}^2 \beta^{t-1} \ln \left(\frac{C_{B,t}^T}{C_{G,t}^T} \right)}_{T \text{ goods}} \\ &= (1 - \omega) \ln \left(\frac{1 + A/Y_2^N}{1 + \Psi A/Y_1^N} \right) - \omega \sum_{t=1}^2 \ln \left(\frac{C_{B,t}^T}{C_{G,t}^T} \right) \end{aligned}$$

where the last equation uses the assumption that $\beta = 1$. Now, notice that in equilibrium, tradable consumption across countries is determined by e_t as follows:

$$C_1^{*T} = (Y_1^T + Y_1^{*T}) \frac{e_1}{1 + e_1} \quad C_1^T = (Y_1^T + Y_1^{*T}) \frac{1}{1 + e_1} \quad (31)$$

$$C_2^{*T} = (Y_2^T + Y_2^{*T}) \frac{e_2 C_2^{*N}}{C_2^N + e_2 C_2^{*N}} \quad C_2^T = (Y_2^T + Y_2^{*T}) \frac{C_2^N}{C_2^N + e_2 C_2^{*N}} \quad (32)$$

The previous equations show that, the higher the exchange rate (stronger dollar), the fewer tradables the EU consumes in equilibrium. Since $e_t^B > e_t^G$, we will have that $C_{B,t}^T < C_{G,t}^T$ for all t . Together with the assumption that $1 + A/Y_2^N > 1 + \Psi A/Y_1^N$, we get $U_G - U_B > 0$.

Equivalently for the US, welfare losses are as follows (assuming $\beta^* = 1$):

$$U_G^* - U_B^* = (1 - \omega) \ln \left(\frac{1 + A^*/Y_2^{*N}}{1 + \Psi^* A^*/Y_1^{*N}} \right) - \omega \sum_{t=1}^2 \ln \left(\frac{C_{B,t}^{*T}}{C_{G,t}^{*T}} \right) \quad (33)$$

Now, from Proposition 1 we have already established that

$$e_1^G = \frac{\eta^*}{\eta + \frac{1}{1+\beta} \frac{1-\omega}{\omega} (A^*/R^* - L^*)} ,$$

while in the case of e_1^B , we get

$$e_1^B = \frac{\eta^*}{\eta - \kappa \cdot \frac{\Omega^* L^* - \Psi^* A^*}{Y_1^{*N} + \Psi^* A^*}} \cdot \frac{Y_1^N + \Psi A}{Y_1^{*N} + \Psi^* A^*} ,$$

where $\kappa \equiv \frac{1-\omega}{\omega} \frac{1}{1+\beta}$. Replacing the values of e_1^G and e_1^B into (31) and (32), and then into (33), combined with the UIP condition $R = R^* \frac{e_2}{e_1}$, yields the inequality in Proposition 5. $\square$

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